

**ROCHESTER INSTITUTE OF TECHNOLOGY  
MICROELECTRONIC ENGINEERING**

# PMOS 150mm PROCESS DETAILS

**Dr. Lynn Fuller, Dr. Ivan Puchades**

Webpage: <http://people.rit.edu/lffeee>

Microelectronic Engineering

Rochester Institute of Technology

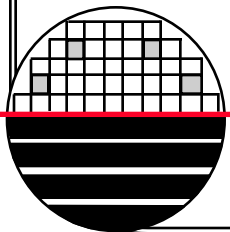
82 Lomb Memorial Drive

Rochester, NY 14623-5604

Tel (585) 475-2035

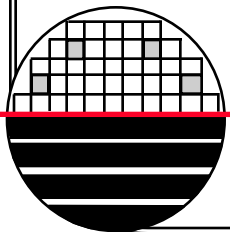
Email: [Lynn.Fuller@rit.edu](mailto:Lynn.Fuller@rit.edu)

MicroE Webpage: <http://www.rit.edu/kgcoe/ue/>



# ***RIT METAL GATE PMOS PROCESS***

- 10 Micrometer Design Rules
- 4 Design Layers
- 4 Photolithography Layers
  - 4 Levels per Plate
- Metal Gate
- Ion Implanted D/S
- Plasma Etched Aluminum
- 150mm Wafer Diameter



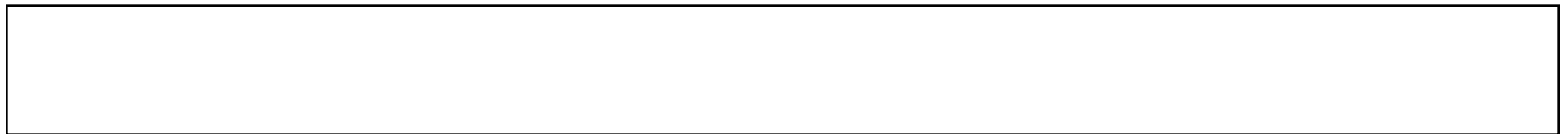
# ***METAL GATE PMOS PROCESS***

## PMOS Versions 150

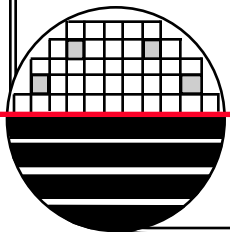
1. ID01 Scribe Wafers C1, C2, D1...DX
2. DE01 4 Point Probe C1, 5-15 ohm-cm
3. CL01 C1, C2, and All Device Wafers
4. OX06 6500Å, Tube 1, Recipe 406
5. PH03 – PMOS Level 1, Diffusion
6. ET06 Step Etch C1, Etch C2 Bare
7. ET06 Etch All Device Wafers
8. ET07 Ash Device Wafers
9. CL01 – RCA Clean All Wafers
- 10.IM01- B11, 75 KeV, All Wafers
- 11.OX06 – 4000Å Tube 1, Recipe 341
- 12.PH03 – PMOS Level 2, Oxide
- 13.ET06 C1, C2 and All Device Wafers
- 14.DE01 4 Point Probe C1
- 15.GRV1 Grove and Stain C1
- 16.ET07 Ash Device Wafers
- 17.CL01 C2 and All Device Wafers, 2 HF dips
- 18.OX05 Gate Ox, Tube 4, 700Å, Recipe 270
- 19.PH03 - PMOS Level 3, CC
- 20.ET06 C2 and All Device Wafers
- 21.ET07 Ash Device Wafers
- 22.DE01 4 pt probe C2
- 23.GRV1 Grove and Stain C2
24. CL01 All Device Wafers, Two HF dips
- 25.ME01 Sputter Metal All Device Wafers 10KÅ
- 26.PH03 - PMOS – Level 4, Metal
- 27.ET15 Etch Aluminum All Device Wafers
- 28.ET07 Ash Device Wafers
- 29.SI01 Sinter Device Wafers Tube 2, Recipe 101
- 30.TE01 Test Wafers



# *STARTING WAFER N-TYPE, 5 OHM-CM*



5-15 ohm-cm, n-type, (100)



## CONTROL WAFERS

C1 (Top Half)

Masking oxide etch rate

Minimum masking oxide thickness

(Bottom Half)

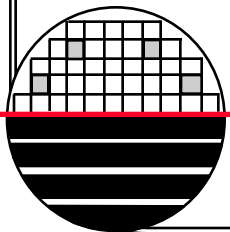
1<sup>st</sup> Junction depth after implant & anneal

Sheet resistance after implant & anneal

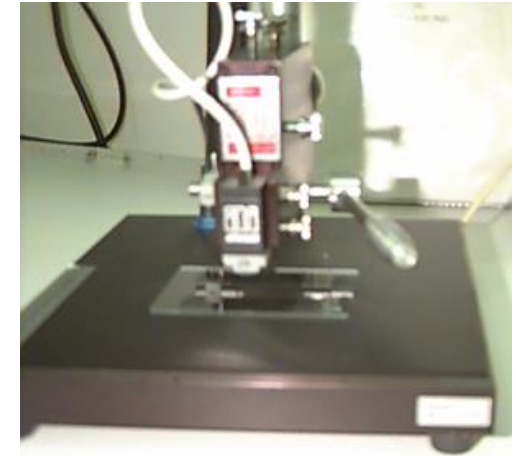
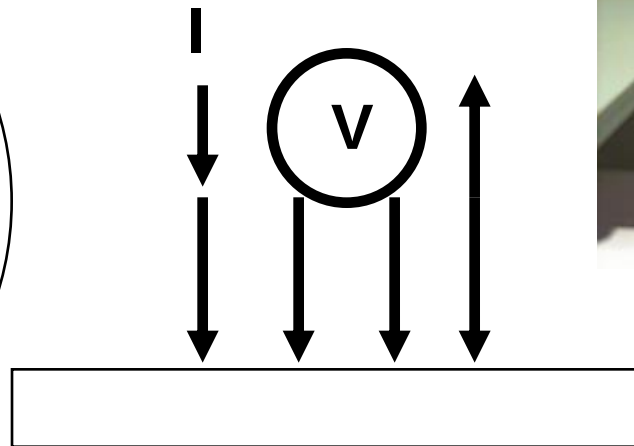
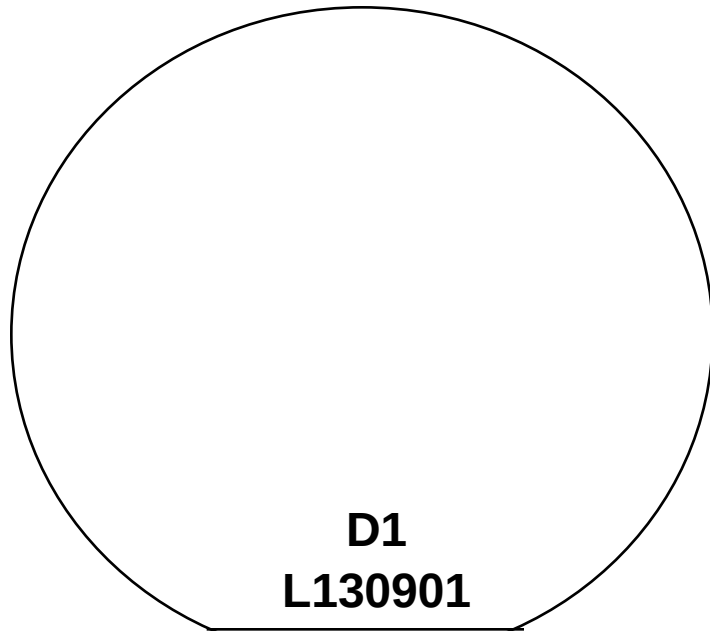
C2

- 2<sup>nd</sup> Junction depth after gate oxide

- Sheet resistance after gate oxide



**ID01 - IDENTIFY WAFER (SCRIBE WAFER)**  
**DE01 - FOUR POINT PROBE**



**S = probe spacing**

**W = wafer thickness**

$$\text{Rho} = \pi / \ln 2 \times W \times V / I \quad \text{ohm-cm}$$

if  $S \ll W$  and  $S \ll \text{Wafer Diameter}$

## DE01 - MEASURE WAFER RESISTIVITY



CDE Resistivity Mapper

CDE ResMap  
RunTitle  
File Name: C:\4P\CDE\_Demo.prj\6in49pt.rcp\3220K051.RaM  
CDE ResMap Demo Recipes

LotID, WaferID: F021111D1 MyWafer  
RunDate: 10:05 02/20/03  
Recip Name: CDE\_Demo 6in49pt  
Oper|Engr|Equip: CDE|Customer [ResMap]

Wafer No. DualPrbCnfg  
WaferDia: 100 Flat  
EdgeExclun: 8.0 FollowMajorFlat

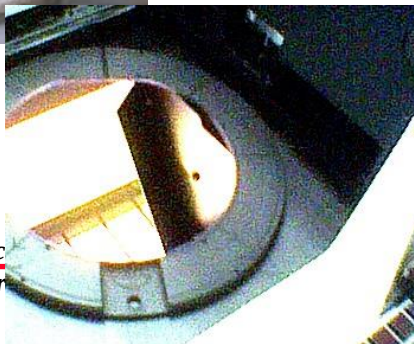
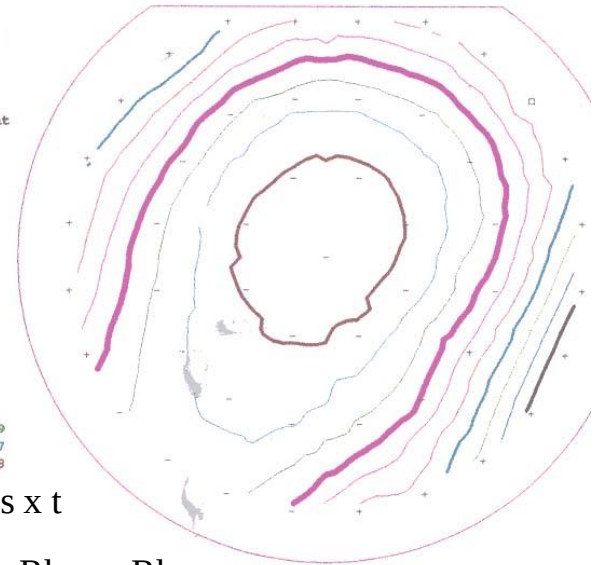
ProbePoints: 49 #Good: 48  
Rs Avg 105.301 Ohms/sq  
StdDev 10.5959 10.063% 3Sgma=30.188%  
Min 90.039 Max 130.19 Range 40.156  
(Mx-Mn)/(Mx+Mn) 18.23% (-)/2Av 19.07%  
Lmin:14.49% Lmax:23.64% (-)/Av 38.13%  
Gradients: R/2=7.652% ~R=22.245%  
Merit: 61.6 21% 42.6 89.6  
Rsns 10.0K IdvMtx 0.74m VanaMtx 14.3m  
DataRejectSigma: 3.0

#data=49 Rs Spacing = 1/3 Sigma

|         |         |
|---------|---------|
| 126.493 | 101.769 |
| 122.961 | 98.2367 |
| 119.429 | 94.7048 |
| 115.897 |         |
| 112.365 |         |
| 108.833 |         |
| 105.301 |         |

$\text{Rho} = \text{Rhos} \times t$

Tool gives Rho or Rhos depending on recipe used, automatically adjusts correction factors for wafer thickness



Record:

Average Resistivity on D1 only

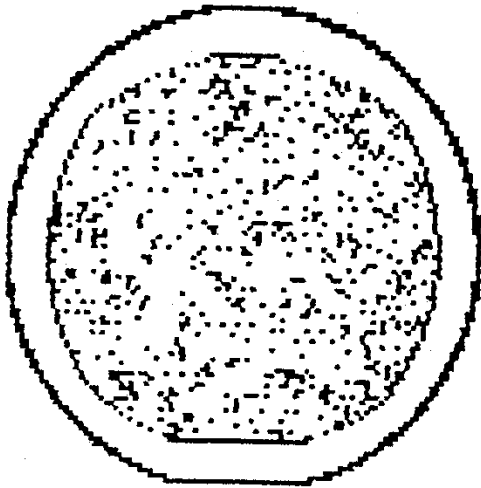
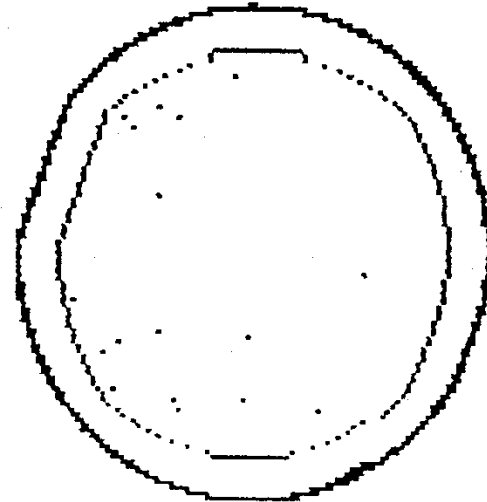
Rho(ave) = ohm-cm

## TENCORE SURF SCAN

Gives total surface particle count and count in 4 bins  $<0.5$ ,  $0.5$  to  $2.0$ ,  $2.0$ - $10$ ,  $>10$ . Bin boundary can be selected. Edge exclusion eliminated count from near the edge of the wafer.





**EXAMPLE SURFACE PARTICLE COUNT DATA****Before Cleaning (75 mm)****Size Range ( $\mu\text{m}$ ) Count****0.2 - 0.5 104****0.5 - 2.0 562****2.0 - 10 19****>10 2****After Cleaning (75 mm)****Size Range ( $\mu\text{m}$ ) Count****0.2 - 0.5 10****0.5 - 2.0 4****2.0 - 10 3****>10 0**

# RCA CLEAN

## APM – SC1

H<sub>2</sub>O – 5000ml  
NH<sub>4</sub>OH – 300ml  
H<sub>2</sub>O<sub>2</sub> – 300ml  
75 °C, 10 min.

DI water  
rinse, 5 min.

H<sub>2</sub>O - 50  
HF - 1  
30 sec.

## HPM – SC2

H<sub>2</sub>O – 5000ml  
HCL – 300ml  
H<sub>2</sub>O<sub>2</sub> – 300ml  
75 °C, 10 min.

DI water  
rinse, 5 min.

DI water  
rinse, 5 min.

SPIN/RINSE  
DRY

What does RCA  
stand for?

**ANSWER**

**PLAY**

## ***RCA CLEAN TOOLS***

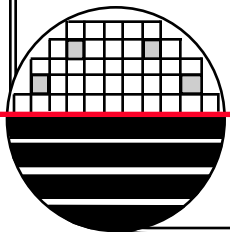
RCA Bench  
Spin/Rinse/Dry Tool



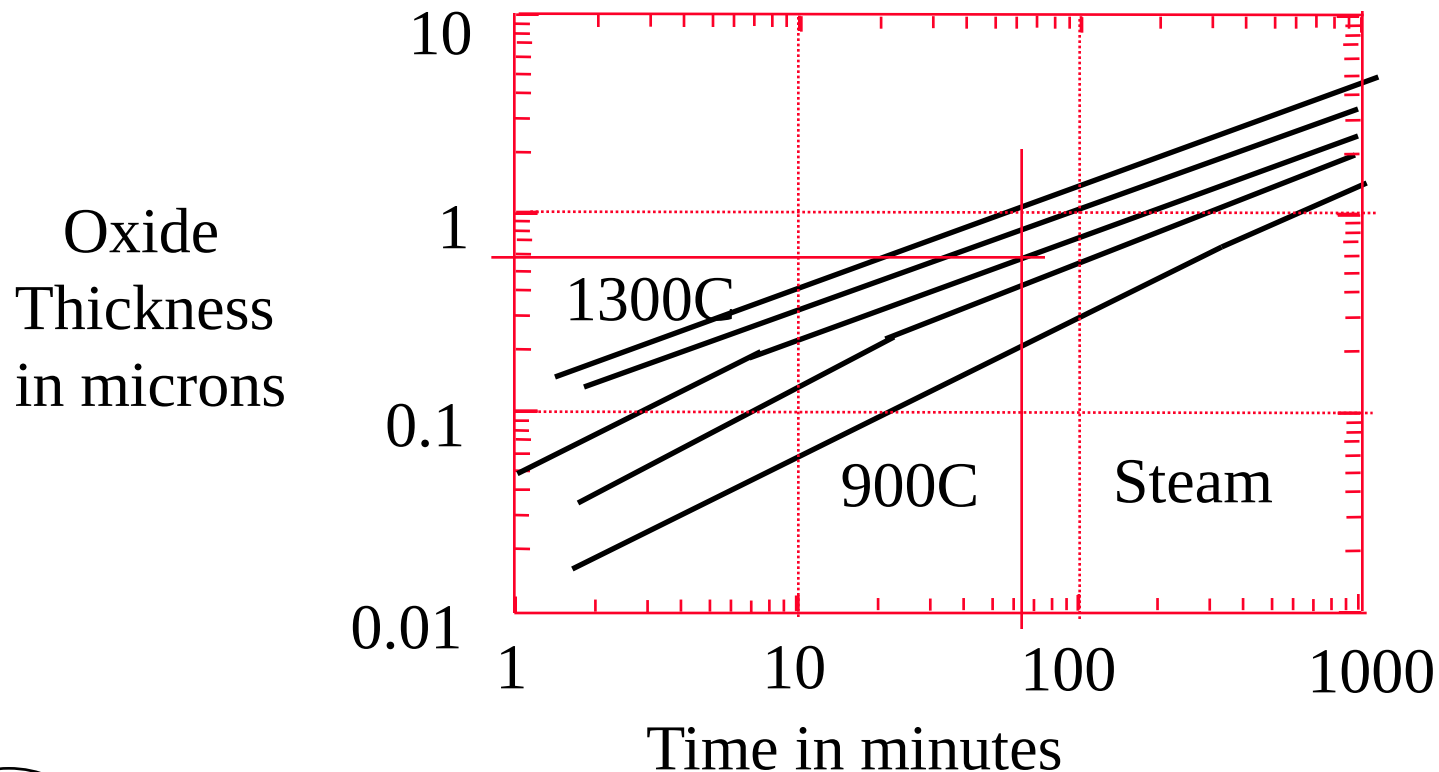
## ***GROW 6500 Å OXIDE***

**6500 Å SiO<sub>2</sub>**

Use Recipe 406 – Tube 1



# WET OXIDE GROWTH CHART



# OXIDE THICKNESS CALCULATOR

ROCHESTER INSTITUTE OF TECHNOLOGY  
MICROELECTRONIC ENGINEERING

OXIDE.XLS  
7/4/2014

CALCULATION OF OXIDE THICKNESS

Dr. Lynn Fuller / Jamie Wasiewicz

To use this spreadsheet change the values in the white boxes. The rest of the sheet is protected and should not be changed unless you are sure of the consequences. The calculated results are shown in the purple boxes. O2 bubbled through warm water is "wet", burning H2 with O2 is called "steam"

| CONSTANTS     |                                    | VARIABLES                                                   | CHOICES                              |
|---------------|------------------------------------|-------------------------------------------------------------|--------------------------------------|
| K             | 1.38E-23 J/K                       |                                                             | 1=yes, 0=no                          |
| (Bo/Ao) dry   | 6230000 $\mu\text{m}/\text{hr}$    | Temp = <input type="text" value="1100"/> $^{\circ}\text{C}$ | wet <input type="text" value="0"/>   |
| Ea (dry)      | 2 eV                               | time = <input type="text" value="65"/> min                  | dry <input type="text" value="0"/>   |
| (Bo/Ao) wet   | 89500000 $\mu\text{m}/\text{hr}$   | Partial Pressure, p = <input type="text" value="1.00"/> Atm | steam <input type="text" value="1"/> |
| Ea (wet)      | 2.05 eV                            |                                                             | <100> <input type="text" value="1"/> |
| (Bo/Ao) steam | 1.63E+08 $\mu\text{m}/\text{hr}$   |                                                             | <111> <input type="text" value="0"/> |
| Ea (steam)    | 2.05 eV                            | Xint= <input type="text" value="0"/> $\text{\AA}$           |                                      |
| Bo dry        | 7.72E+02 $\mu\text{m}^2/\text{hr}$ |                                                             |                                      |
| Ea (dry)      | 1.23 eV                            |                                                             |                                      |
| Bo wet        | 2.14E+02 $\mu\text{m}^2/\text{hr}$ | Silicon VLSI Technology, Phummer, Deal, Griffin             |                                      |
| Ea (wet)      | 0.71 eV                            | Prentice Hall, 2000, pg 319-369                             |                                      |
| Bo steam      | 3.86E+02 $\mu\text{m}^2/\text{hr}$ |                                                             |                                      |
| Ea (steam)    | 0.78 eV                            | (Bo/Ao)/1.68 for <100>                                      |                                      |

CALCULATIONS:

$$X_{ox} (\text{Oxide thickness}) = (A/2) \{ [1 + (t + \tau) 4B/A^2]^{\frac{1}{2}} - 1 \} = 6765 \text{ \AA}$$

$$B = [Bo \exp(-Ea/KTemp)] * p$$

$$B/A = [(Bo/Ao) \exp(-Ea/KTemp)] * p$$

$$A =$$

$$\tau = (Xi^2 + AXi)/B$$

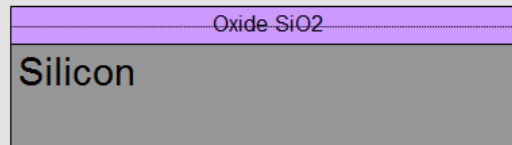
$$0.5338344 \mu\text{m}^2/\text{hr}$$

$$2.99E+00 \mu\text{m}/\text{hr}$$

$$0.1784766 \mu\text{m}$$

$$0 \text{ hr}$$

X<sub>ox</sub>



Original Silicon Surface Prior to Oxide Growth

0.46 X<sub>ox</sub> (silicon consumed)

1100°C

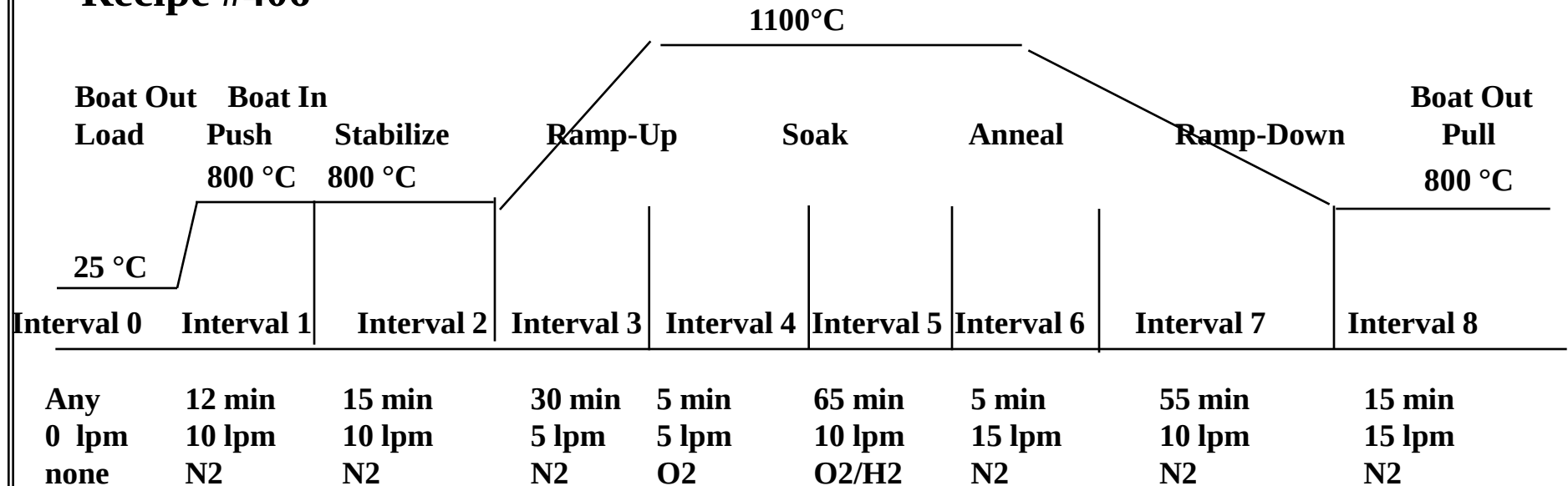
65 min

Steam

6765 Å

# BRUCE FURNACE RECIPE 406 – WET OXIDE 6,500Å

## Recipe #406



At the end of a run the furnace returns to Interval 0 which is set for boat out, 25 °C and no gas flow. The furnace waits in that state until someone aborts the current recipe or loads a new recipe.

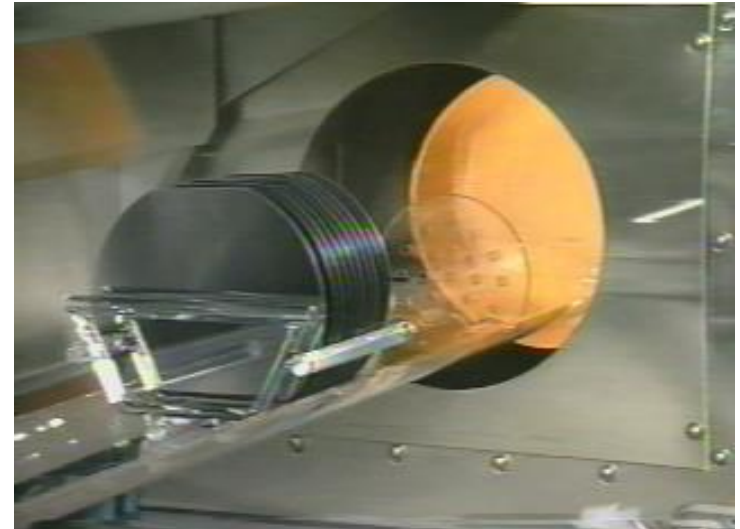
## Wet Oxide Growth, Target 6,500 Å

Rochester Institute of Technology  
Microelectronic Engineering

## 6500 Å OXIDE GROWTH

### PMOSFET

1100°C  
65 min  
Steam  
Bruce Tube 1



n-type substrate, 5-15 ohm-cm, (100)

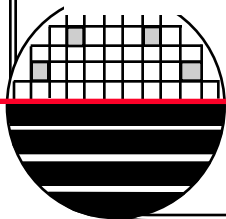


## OXIDE COLOR VERSUS THICKNESS TABLE

| Thickness | Color                                    |
|-----------|------------------------------------------|
| 500       | Tan                                      |
| 700       | Brown                                    |
| 1000      | Dark Violet - Red Violet                 |
| 1200      | Royal Blue                               |
| 1500      | Light Blue - Metallic Blue               |
| 1700      | Metallic - very light Yellow Green       |
| 2000      | Light Gold or Yellow - Slightly Metallic |
| 2200      | Gold with slight Yellow Orange           |
| 2500      | Orange - Melon                           |
| 2700      | Red Violet                               |
| 3000      | Blue - Violet Blue                       |
| 3100      | Blue                                     |
| 3200      | Blue - Blue Green                        |
| 3400      | Light Green                              |
| 3500      | Green - Yellow Green                     |
| 3600      | Yellow Green                             |
| 3700      | Yellow                                   |
| 3900      | Light Orange                             |
| 4100      | Carnation Pink                           |
| 4200      | Violet Red                               |
| 4400      | Red Violet                               |
| 4600      | Violet                                   |
| 4700      | Blue Violet                              |

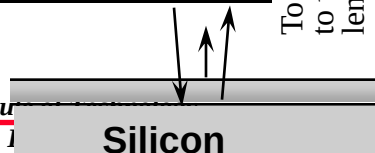
To observe a valid color, the wafer must be observed perpendicular to the surface under white (all wavelengths) light or the optical path length will be different, hence the color will change with the angle.

| Thickness | Color                                                            |
|-----------|------------------------------------------------------------------|
| 4900      | Blue                                                             |
| 5000      | Blue Green                                                       |
| 5200      | Green                                                            |
| 5400      | Yellow Green                                                     |
| 5600      | GreenYellow                                                      |
| 5700      | Yellow - "Yellowish"(at times appears to be Lt gray or metallic) |
| 5800      | Light Orange or Yellow - Pink                                    |
| 6000      | Carnation Pink                                                   |
| 6300      | Violet Red                                                       |
| 6800      | "Bluish"(appears violet red, Blue Green, looks grayish)          |
| 7200      | Blue Green - Green                                               |
| 7700      | "Yellowish"                                                      |
| 8000      | Orange                                                           |
| 8200      | Salmon                                                           |
| 8500      | Dull, Light Red Violet                                           |
| 8600      | Violet                                                           |
| 8700      | Blue Violet                                                      |
| 8900      | Blue                                                             |
| 9200      | Blue Green                                                       |
| 9500      | Dull Yellow Green                                                |
| 9700      | Yellow - "Yellowish"                                             |
| 9900      | Orange                                                           |
| 10000     | Carnation Pink                                                   |



**Yes!**

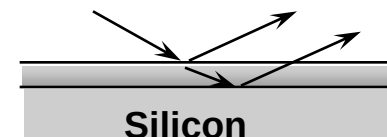
Rochester Institute of Technology  
Microelectronics



$\text{SiO}_2$

Silicon

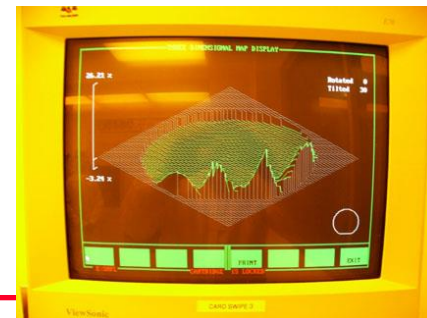
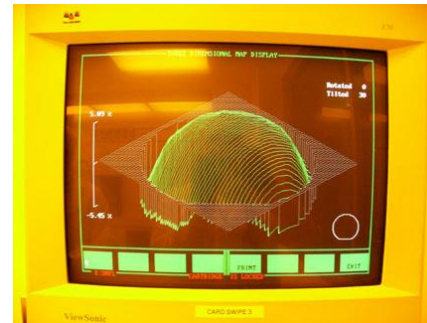
**No!**



$\text{SiO}_2$

Silicon

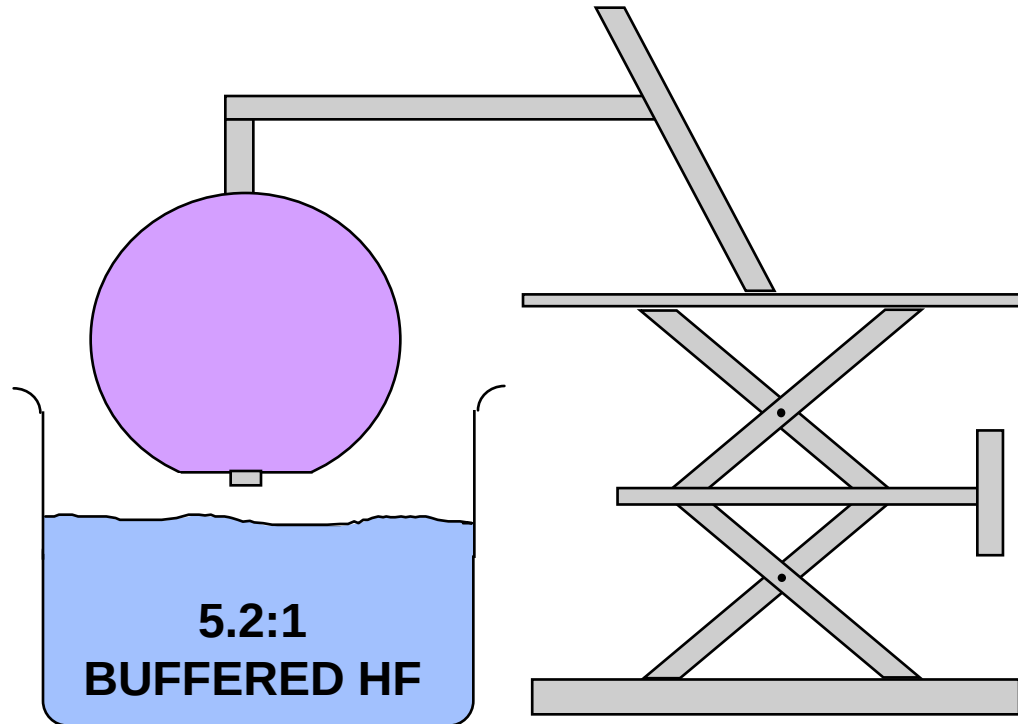
# TENCORE SPECROMAP



Record:

Mean  
Std Deviation  
Min  
Max  
No of Points

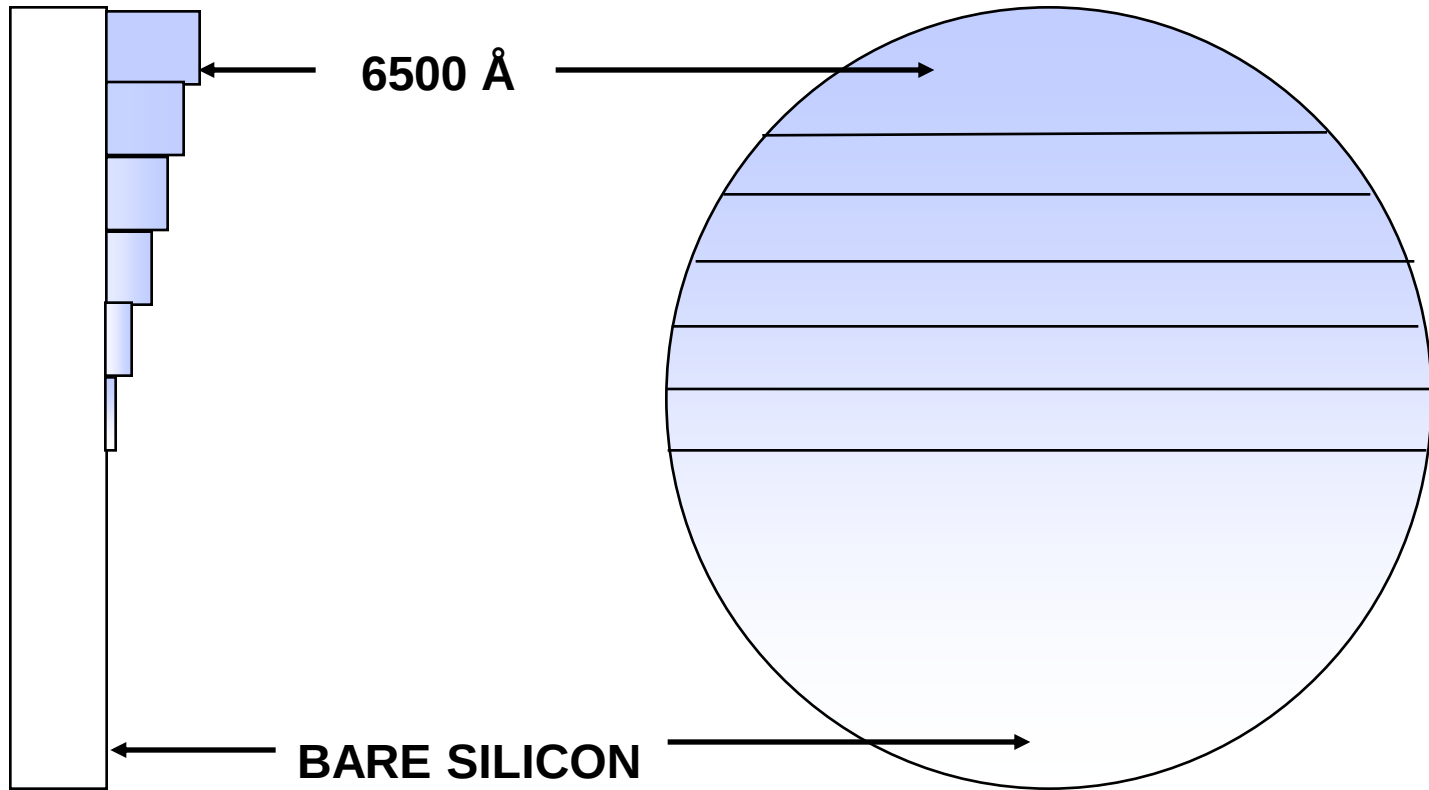
## STEP ETCH APPARATUS



**Lower 1/4 inch every 30 seconds**



## *ETCH STEPS IN OXIDE ON C1*

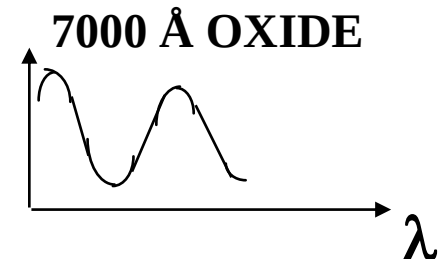
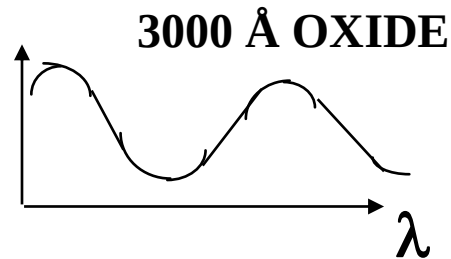
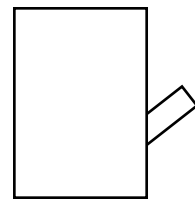


# REFLECTANCE SPECTROMETER

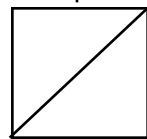
## NANOSPEC FILM THICKNESS MEASUREMENT

INCIDENT WHITE LIGHT, THE INTENSITY OF THE REFLECTED LIGHT IS MEASURED VS WAVELENGTH

MONOCHROMATOR  
& DETECTOR



WHITE LIGHT SOURCE



OPTICS

WAFER



|                             |              |
|-----------------------------|--------------|
| Oxide on Silicon            | 400-30,000 Å |
| Nitride                     | 400-30,000   |
| Neg Resist                  | 500-40,000   |
| Poly on 300-1200 Ox         | 400-10,000   |
| Neg Resist on Ox 300-350    | 300-3500     |
| Nitride on Oxide 300-3500   | 300-3500     |
| Thin Oxide                  | 100-500      |
| Thin Nitride                | 100-500      |
| Polyimide                   | 500-10,000   |
| Positive Resist             | 500-40,000   |
| Pos Resist on Ox 500-15,000 | 4,000-30,000 |

## MEASURE OXIDE THICKNESS



Record:

Color =

Color Chart Thickness =      Å

Nanospec Thickness =      Å

## ***COAT WITH PHOTORESIST***

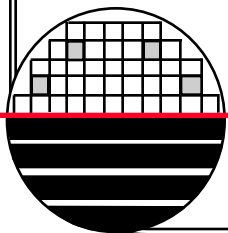
**1  $\mu\text{m}$  Photoresist**

**6500 Å SiO<sub>2</sub>**

SSI Coat Recipe - COAT  
SSI Develop Recipe - DEVELOP  
Stepper Job Name - PMOS

# LITHOGRAPHY FOR PMOS 150 PROCESS

| Lv<br>l # | Level<br>Name | Coat<br>Recipe | Spin<br>RPM | Xpr<br>$\mu\text{m}$ | Dose<br>$\text{mj}/\text{cm}^2$ | Dev<br>Recipe | Dev<br>Time | Hard Bake |
|-----------|---------------|----------------|-------------|----------------------|---------------------------------|---------------|-------------|-----------|
| 1         | Diff          | coat           | 3250        | 1.0                  | 250                             | develop       | 50s         | 140C/1min |
| 2         | ThinOx        | coatmtl        | 2000        | 1.3                  | 250                             | devmtl        | 68s         | 140C/2min |
| 3         | CC            | coatmtl        | 2000        | 1.3                  | 250                             | devCC         | 180s        | 140C/1min |
| 4         | Metal         | coatmtl        | 2000        | 1.3                  | 250                             | devmtl        | 68s         | 140C/2min |





## COAT.RCP

DEHYDRATE BAKE/  
HMDS PRIMING

HMDS Vapor  
Prime  
140 °C, 60 sec.

COAT.RCP  
SPIN COAT

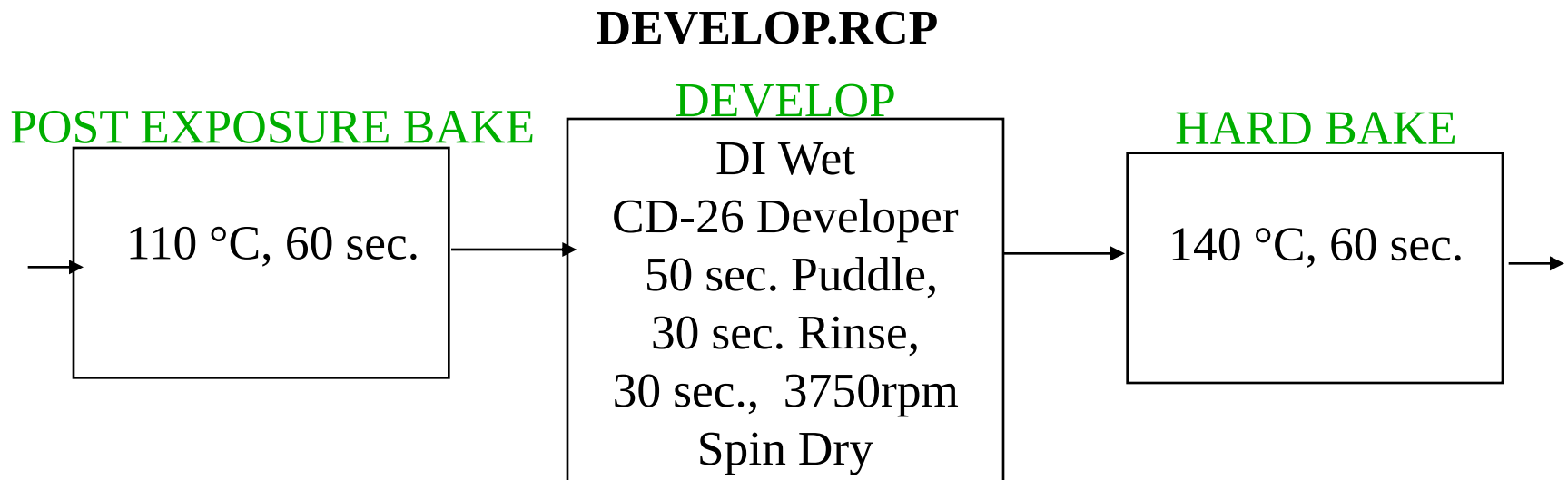
OIR 620-10  
Resist  
3250 rpm, 30 sec.

SOFT BAKE

90 °C  
60 sec.

Thickness of 10,000 Å

## ***DEVELOP.RCP***



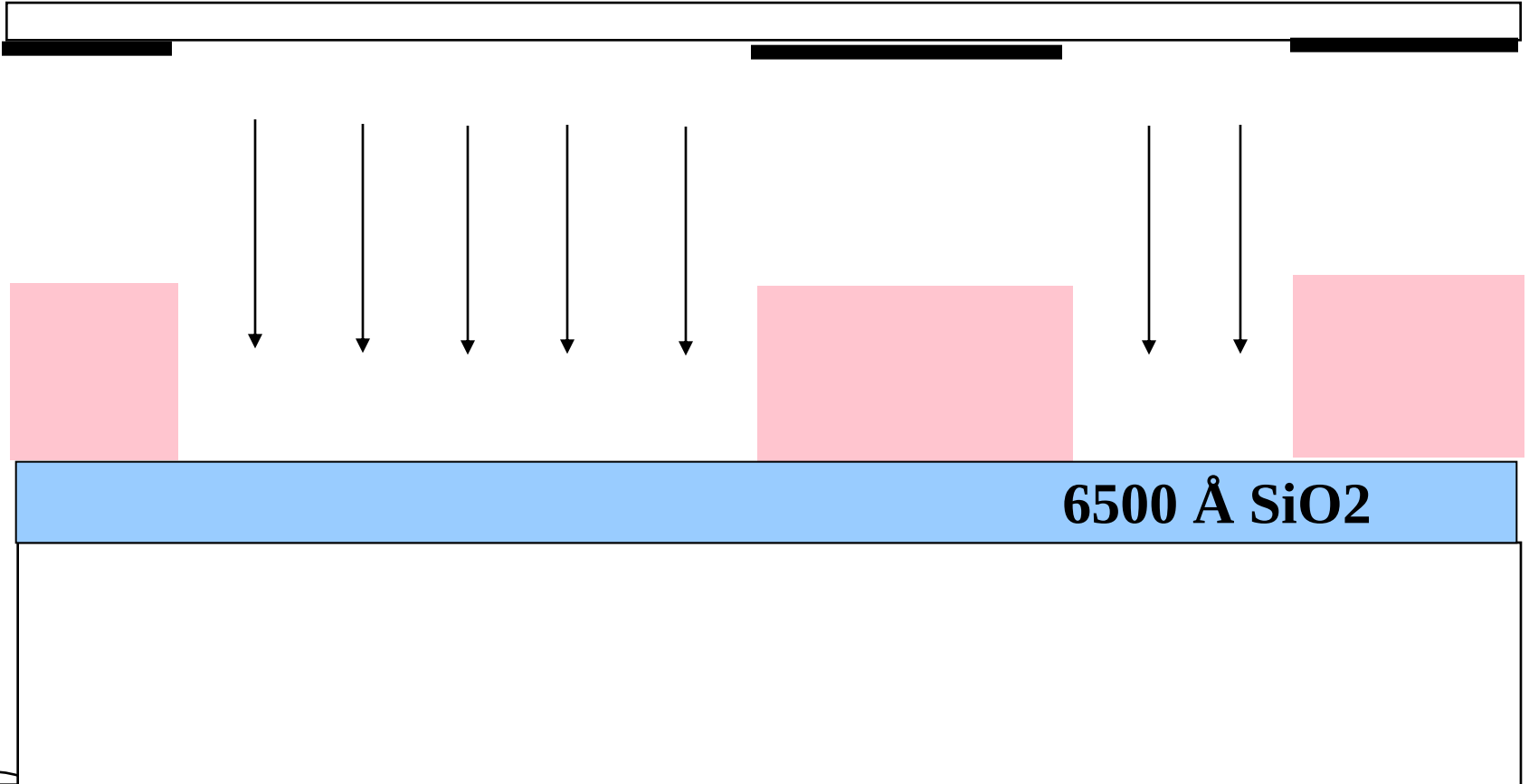
## *SSI COAT AND DEVELOP TRACK FOR 6" WAFERS*



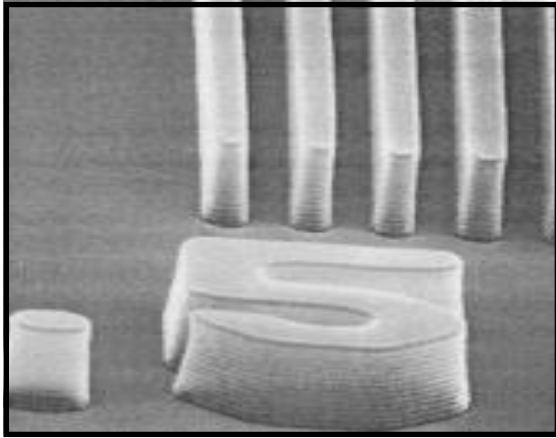
Use Recipe: Coat.rcp and Develop.rcp

*Rochester Institute of Technology*  
*Microelectronic Engineering*

***EXPOSE WITH LEVEL ONE DIFFUSION***



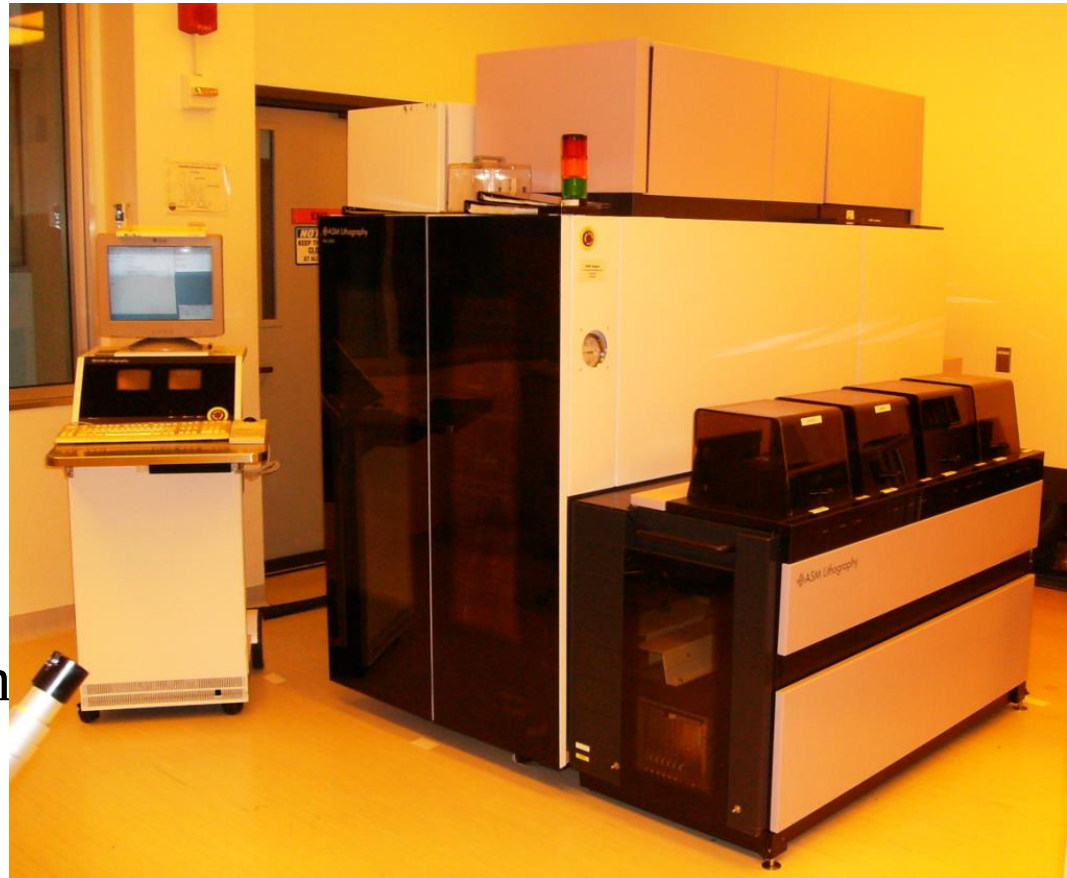
# ASML 5500/200 STEPPER



NA = 0.48 to 0.60 variable  
 $\sigma = 0.35$  to 0.85 variable  
 With Variable Kohler, or  
 Variable Annular illumination  
 Resolution =  $K_1 \lambda / NA$   
 $\approx 0.35 \mu m$

for NA=0.6,  $\sigma = 0.85$

Depth of Focus =  $k_2 \lambda / (NA)^2$   
 $\approx 1.0 \mu m$  for NA = 0.6



i-Line Stepper  $\lambda = 365 \text{ nm}$   
 22 x 27 mm Field Size

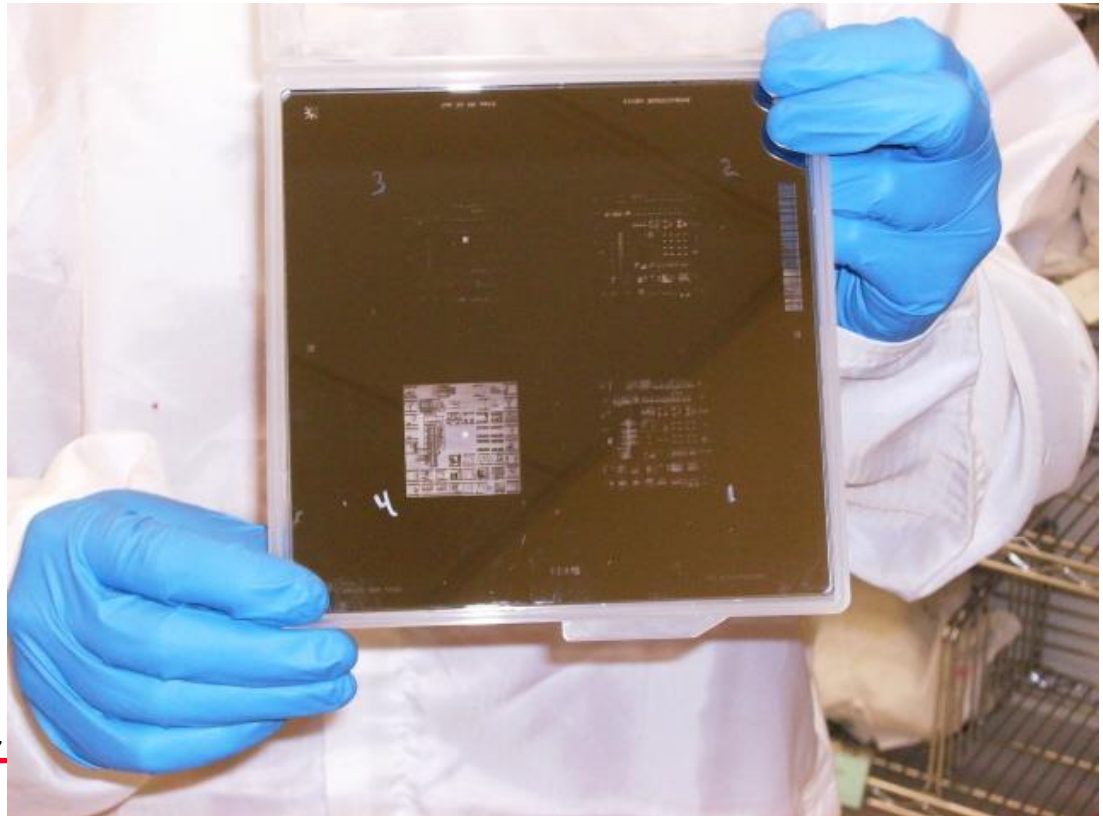
## ***MASKS AND STEPPER JOBS***

Masks with 4 levels

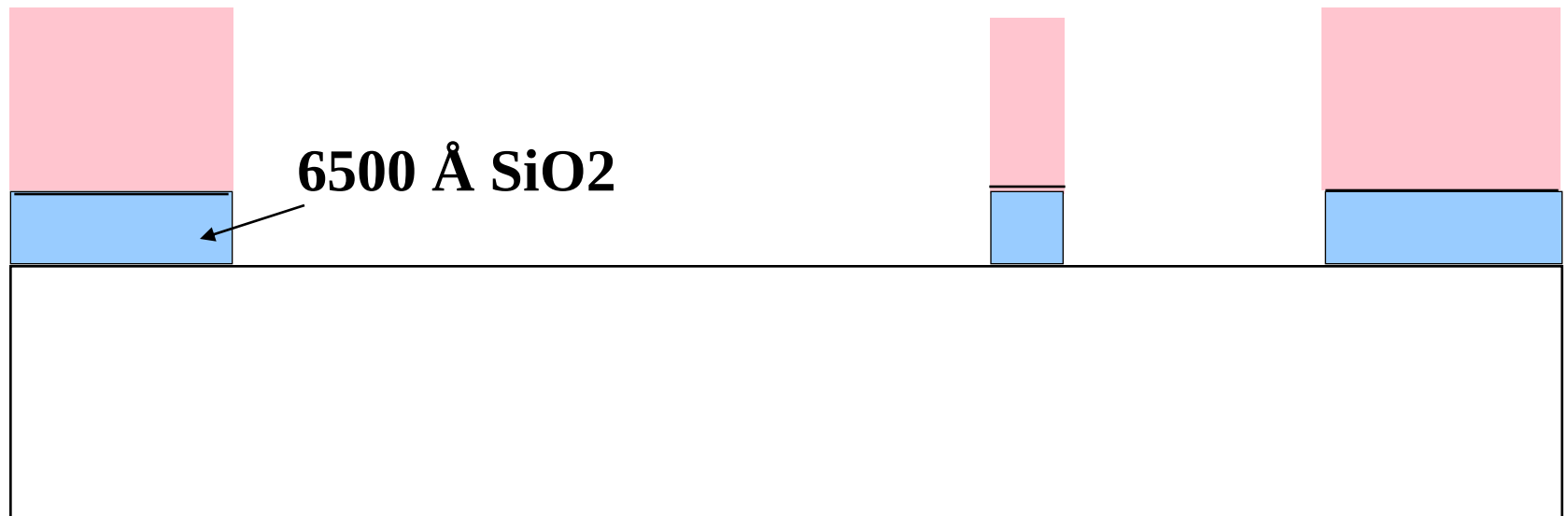
Saves money, time, inventory

Chip size 10mm by 10mm

Stepper Job Name = PMOS



## *ETCH OXIDE*

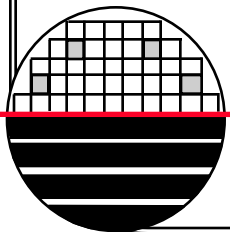


Buffered HF

If 5.2:1 BOE time = 8 min

If 10:1 BOE time = 15 min

Rinse/Spin Dry





## ASHER, RCA CLEAN & SRD



O is reactive and will combine with plastics, wood, carbon, photoresist, etc.

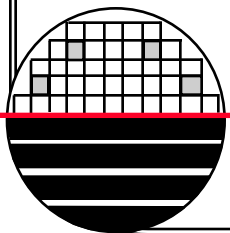


Gassonics Asher

*Rochester Institute of Technology  
Microelectronic Engineering*



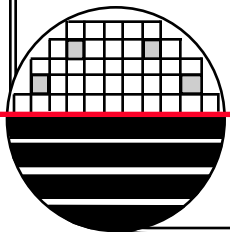
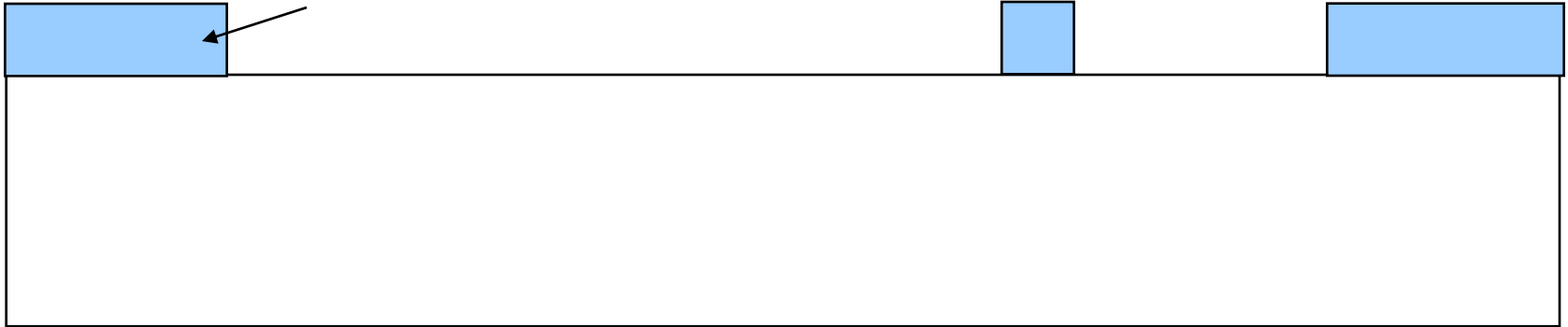
RCA Clean Bench





## *AFTER ASH RESIST AND CLEAN*

6500 Å SiO<sub>2</sub>

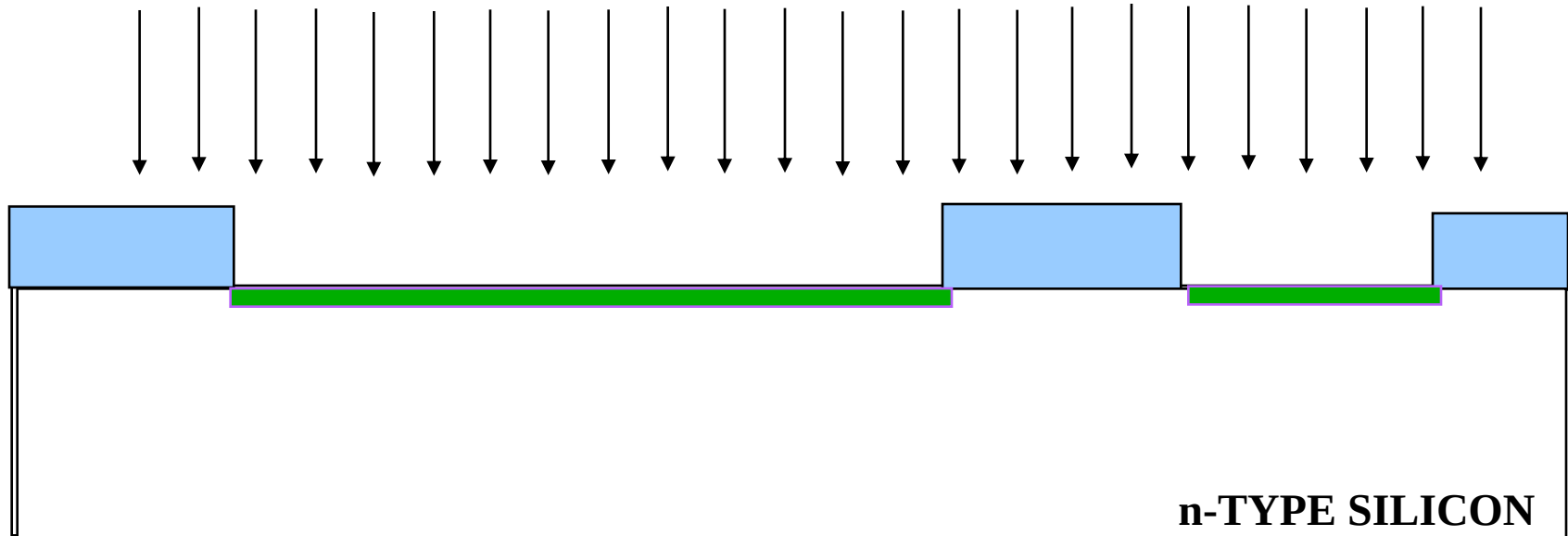


## *VARIAN 350 D ION IMPLANTER (4" AND 6" WAFERS)*



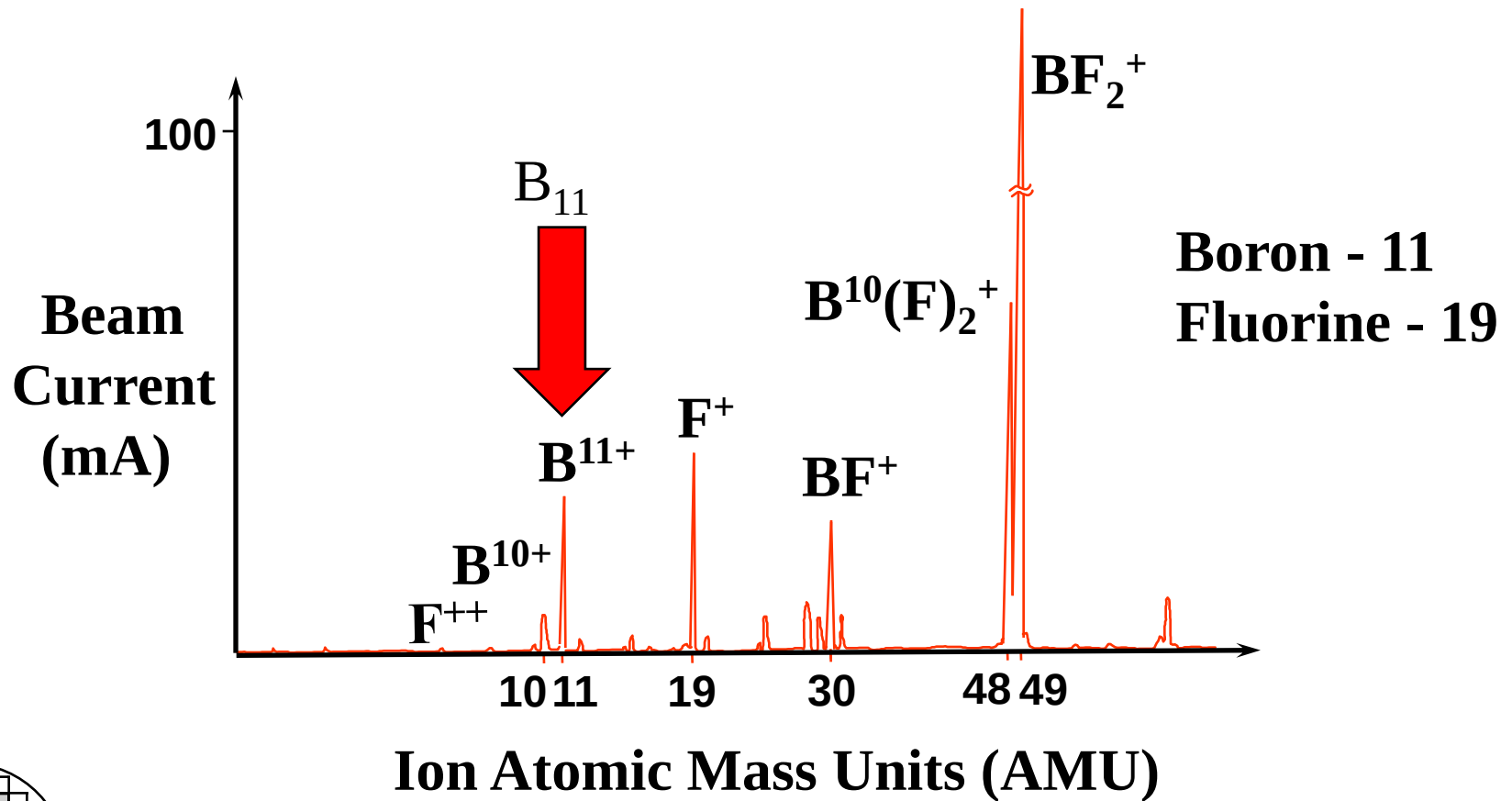
*Rochester Institute of Technology  
Microelectronic Engineering*

## ION IMPLANT P-TYPE DOPANT



All device wafers  
All control wafers

Boron  
Species: B<sub>11</sub>  
Gas: BF<sub>3</sub>  
Dose:  $2 \times 10^{15}$  ions/cm<sup>2</sup>  
Energy: 75 KeV  
Implant Time: ~ 5 minutes per wafer  
@225μA

***BF<sub>3</sub> GAS SPECTRUM***

# IMPLANT MASKING THICKNESS CALCULATOR

Rochester Institute of Technology

Microelectronic Engineering

11/20/2004

Lance Barron

Dr. Lynn Fuller

## IMPLANT MASK CALCULATOR

Enter 1 - Yes 0 - No in white boxes

### DOPANT SPECIES

B11

1

BF2

0

P31

0

### MASK TYPE

Resist

0

Poly

0

Oxide

1

Nitride

0

### ENERGY

75

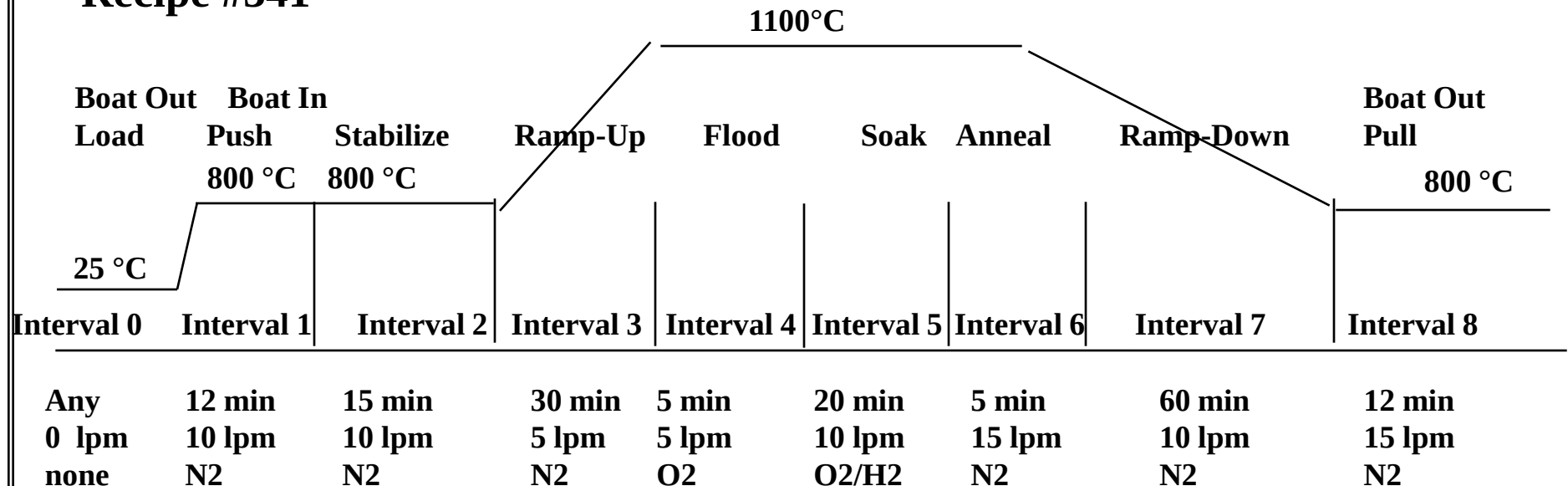
KeV

Thickness to Mask  $>1E15/cm^3$  Surface Concentration

3971.104 Angstroms

# BRUCE FURNACE RECIPE 341 – WET OXIDE 4,000Å

## Recipe #341

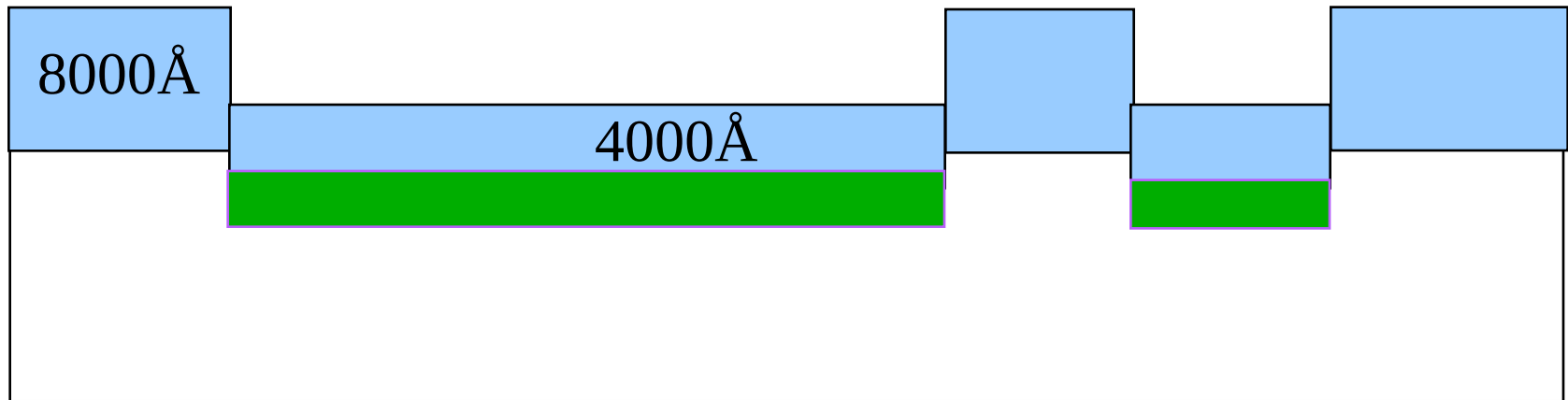


At the end of a run the furnace returns to Interval 0 which is set for boat out, 25 °C and no gas flow. The furnace waits in that state until someone aborts the current recipe or loads a new recipe.

## Wet Oxide Growth, Target 4000 Å

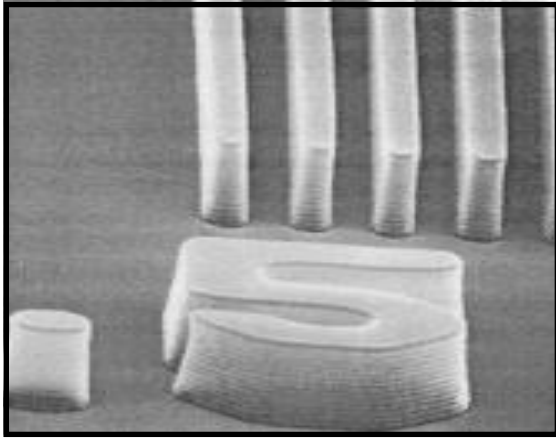
Rochester Institute of Technology  
Microelectronic Engineering

## AFTER ANNEAL AND OXIDE GROWTH



Step height in silicon =  $2000\text{\AA} - \frac{1}{2} 1500\text{\AA} = 1250\text{\AA}$   
Important for ASML Overlay

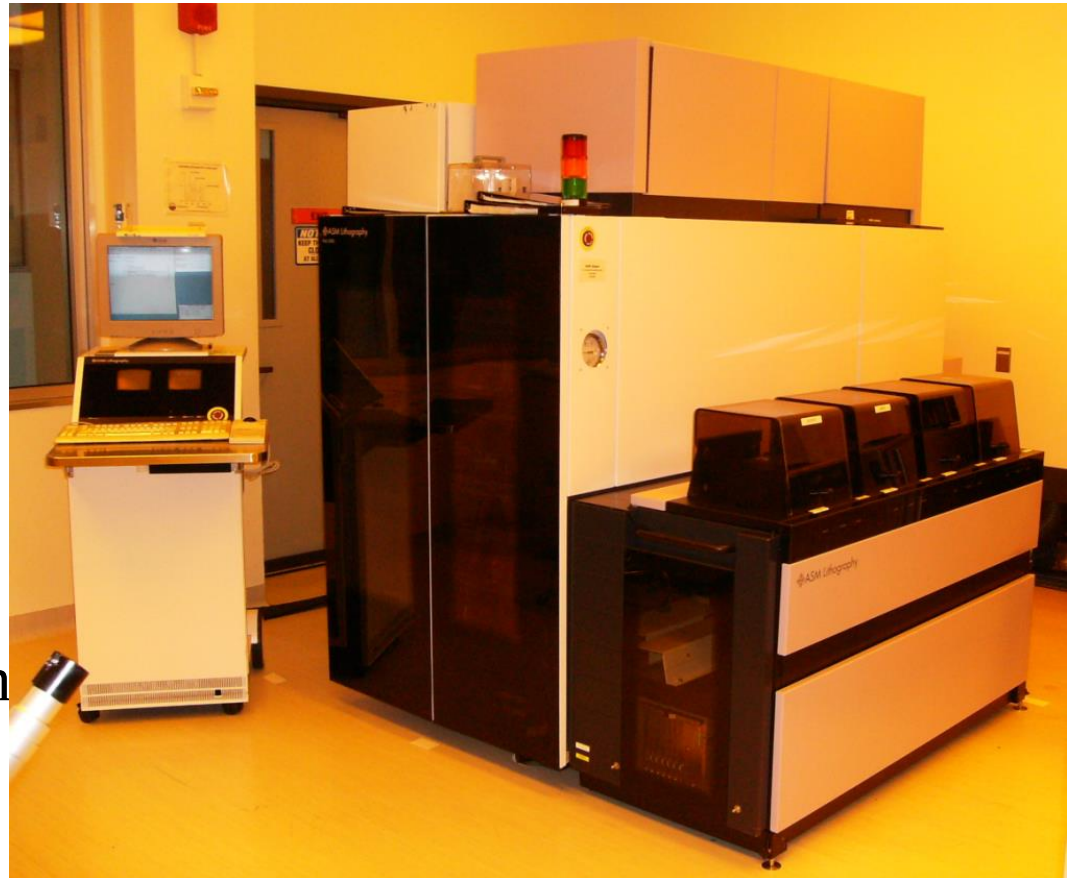
# ASML 5500/200 STEPPER



NA = 0.48 to 0.60 variable  
 $\sigma = 0.35$  to 0.85 variable  
 With Variable Kohler, or  
 Variable Annular illumination  
 Resolution =  $K_1 \lambda / NA$   
 $\approx 0.35 \mu m$

for NA=0.6,  $\sigma = 0.85$

Depth of Focus =  $k_2 \lambda / (NA)^2$   
 $\approx 1.0 \mu m$  for NA = 0.6



i-Line Stepper  $\lambda = 365 \text{ nm}$   
 22 x 27 mm Field Size

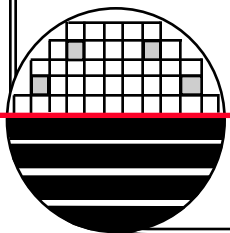


## ALIGNMENT

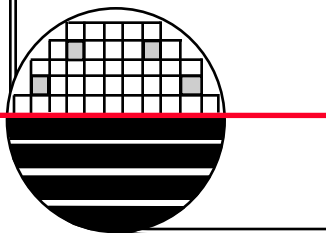
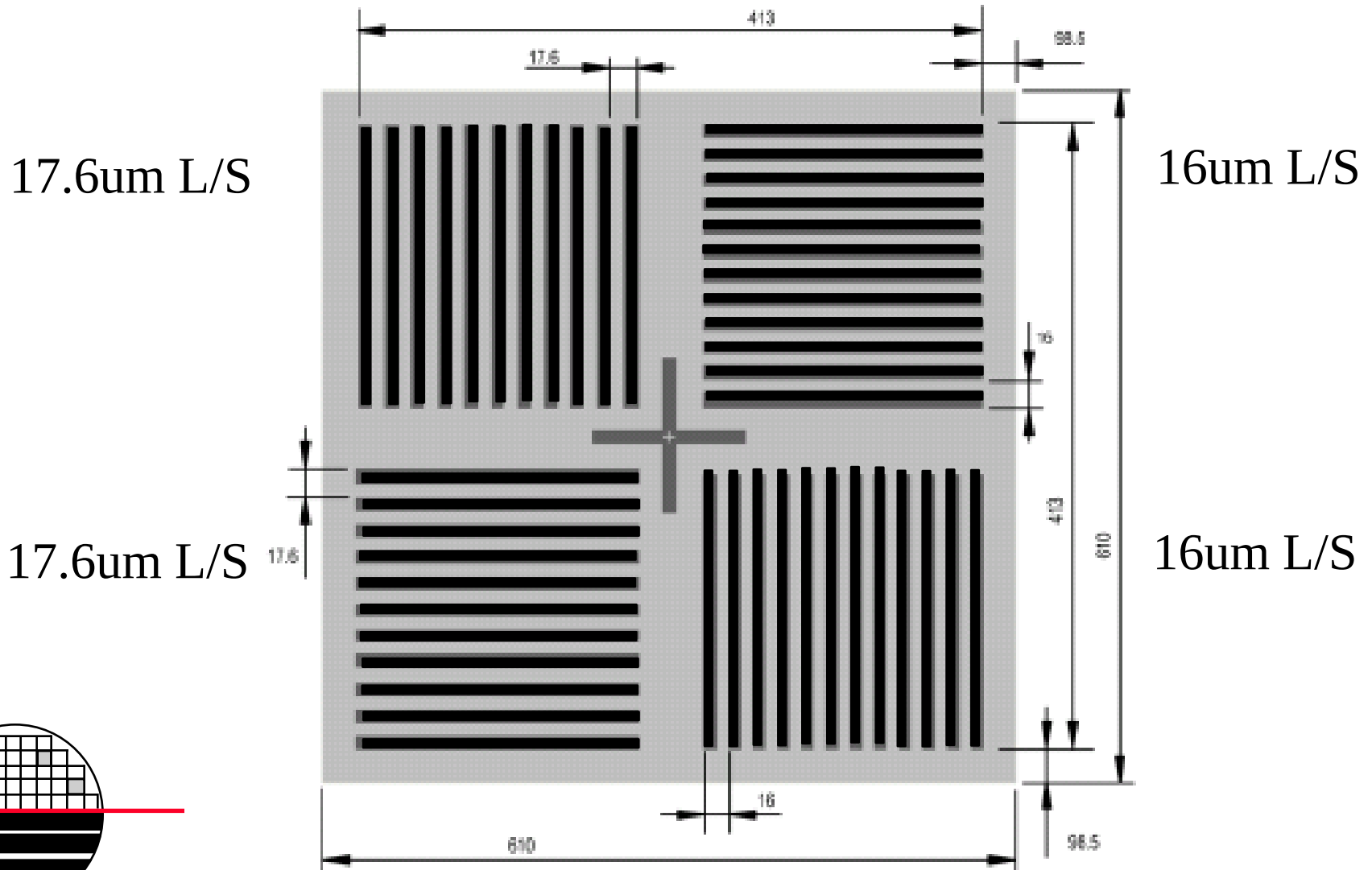
The ASML PAS 5500 uses wafer alignment marks that are diffraction gratings. There are marks for both the x and y directions. These marks are illuminated with a HeNe laser at a single wavelength near 632.8nm. The reflected wave exhibits a diffraction pattern of bright and dark lines that are focused on a sensor. The stage is moved slightly to learn the best position to match the sensor and that stage position is used to calculate the stage position to place the die under the center of the optical column. The wafer is moved to the lens center (or shifted by a fixed amount from center) and the die is exposed. The stage position for the remaining die are calculated and those die are also exposed. The wafer marks are lines and spaces etched into the starting wafer. To give maximum contrast in the diffracted pattern the etch depth  $\lambda/4n$  results in a optical path difference of  $\pi$ ,  $\lambda$  is the wavelength of the laser light and  $n$  is the index of refraction of the material above the marks (usually photoresist or oxide). The etch depth calculation gives a value of approximately  $632.8/4/1.45 = 110 \text{ nm}$  (1100Å)

## ALIGNMENT

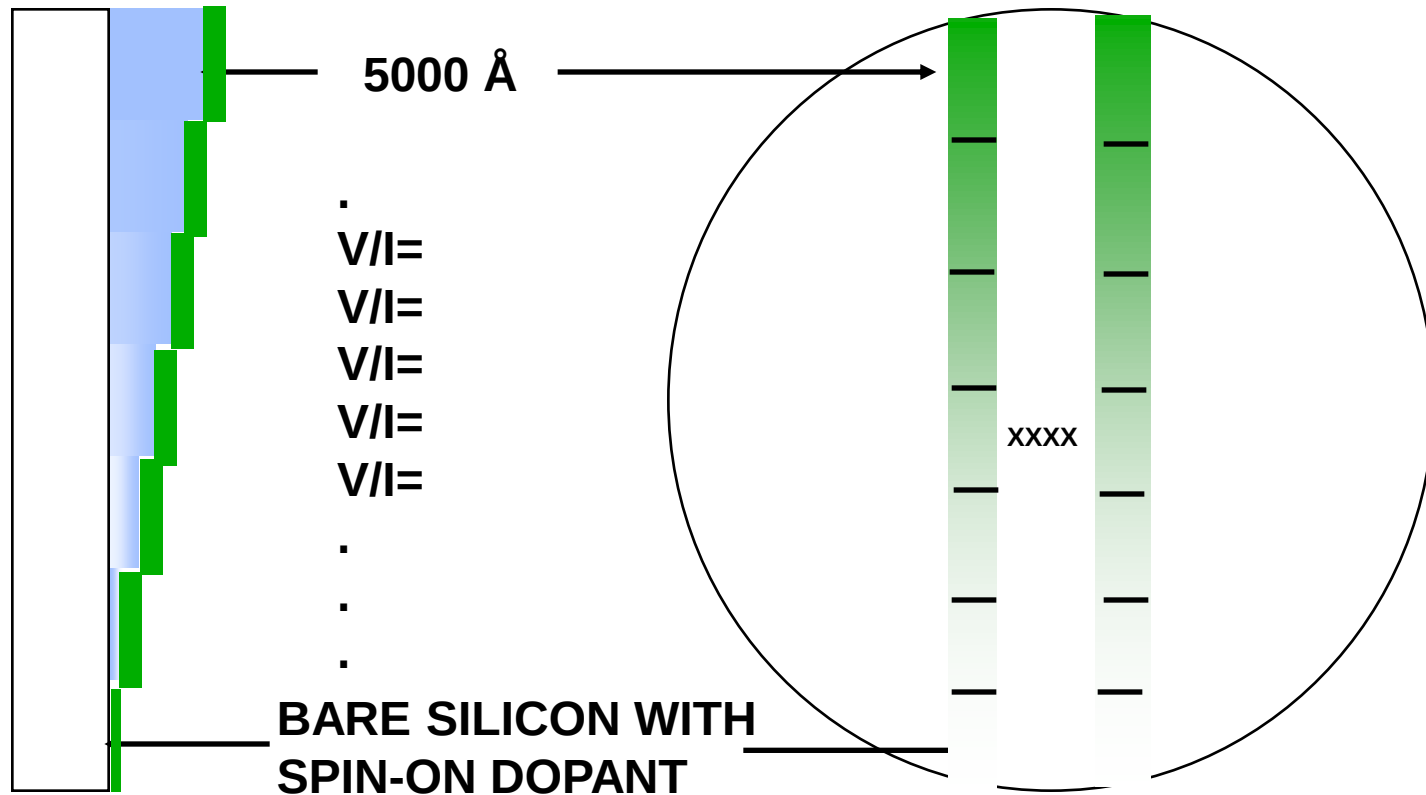
Alignment involves placing the wafer /stage in a position such that the wafer/stage marks can be illuminated by the HeNe laser. The reflected diffraction pattern goes back through the lens and the wafer image is reconstructed from the  $\pm 1^{\text{st}}$  order components of the diffraction pattern (the zero order is returned to the laser, higher orders are blocked). The electric and magnetic fields are transferred through the lens as in a linear system resulting in a sinusoidal field image. The intensity is the square of the field doubling the frequency of the diffraction grating on the wafer when viewed at the mask level. This image is superimposed on the fiducial marks on the reticle and a light detector measures the brightness as the stage is moved to find best alignment of the wafer to the mask.



# ANIMATION OF WAFER ALIGNMENT TO RETICLE



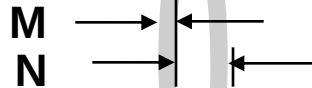
# PAINT RESIST STRIP ETCH C1 BARE



**FIND MINIMUM OXIDE THICKNESS TO MASK BORON IMPLANT**

# GROOVE and STAIN C1 FIND $X_j$ AFTER PRE DEPOSIT

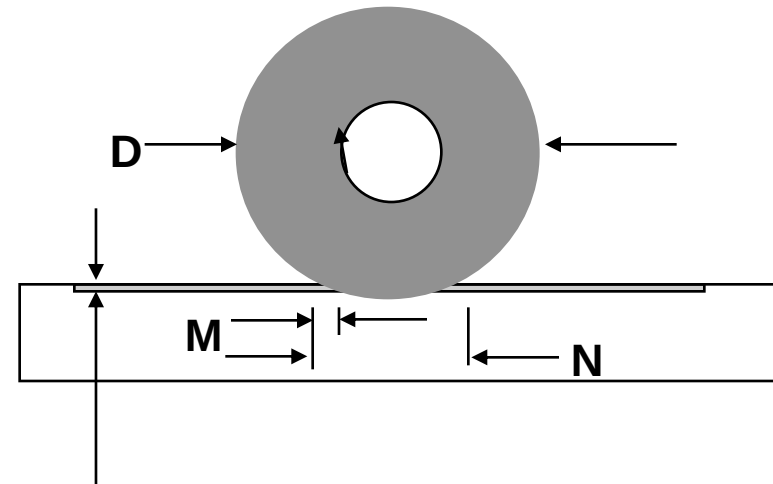
M  
N



After Stain



Groove



$$X_j = (N * M) / D$$

(at RIT D=1.532 inch)

Staining Solution - 1 Vol part HF, 2 Vol part Nitric Acid, 12 Vol part Acetic Acid  
After mixing drop a penny in solution for about 10 sec. result in a light blue color. Safety Stain - (does not have HF) is available from Philtec Instrument Co. Philadelphia, PA 19129-1651, (215) 848-4500, Signatone makes groove tool and wheels, (408)732-3280

Rochester Institute of Technology  
Microelectronic Engineering

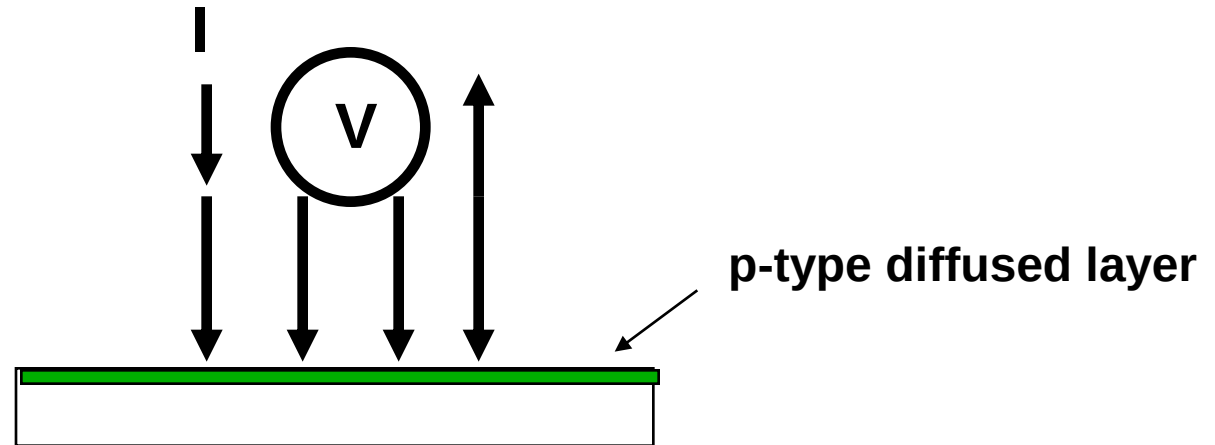
# TRAVELING STAGE MICROSCOPE



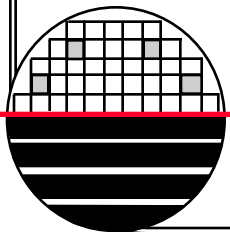
Example: If  $M=0.003$  inches  
and  $N=0.025$  inches, find  $x_j$ .

$$X_j = (N * M) / D$$

$$\begin{aligned} &= (0.025 * 0.003) / 1.588 \text{ inch} \\ &= 0.0000472 \text{ inch} \\ &= 1.20 \mu\text{m} \end{aligned}$$

**DE01 - FOUR POINT PROBE BOTTOM OF C1****FIND SHEET RESISTANCE OF IMPLANT P+**

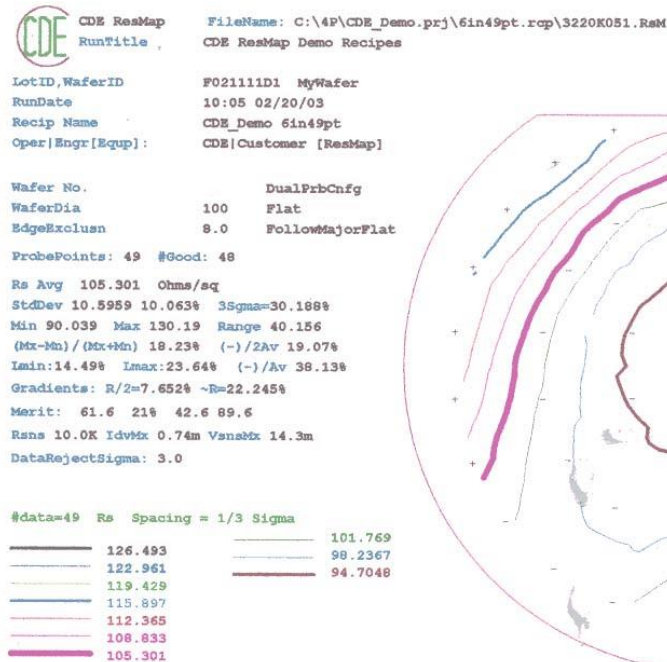
$$R_{\text{hos}} = V / I * \pi / \ln 2 \text{ ohms/square} = 4.53 V / I \text{ ohms/sq}$$



## MEASURE SHEET RESISTANCE

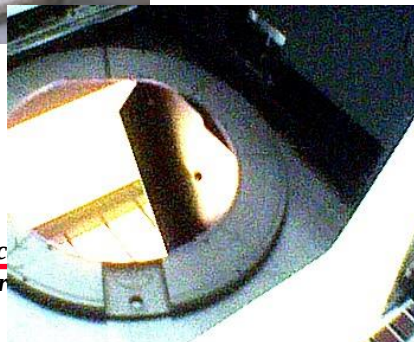


CDE Resistivity Mapper



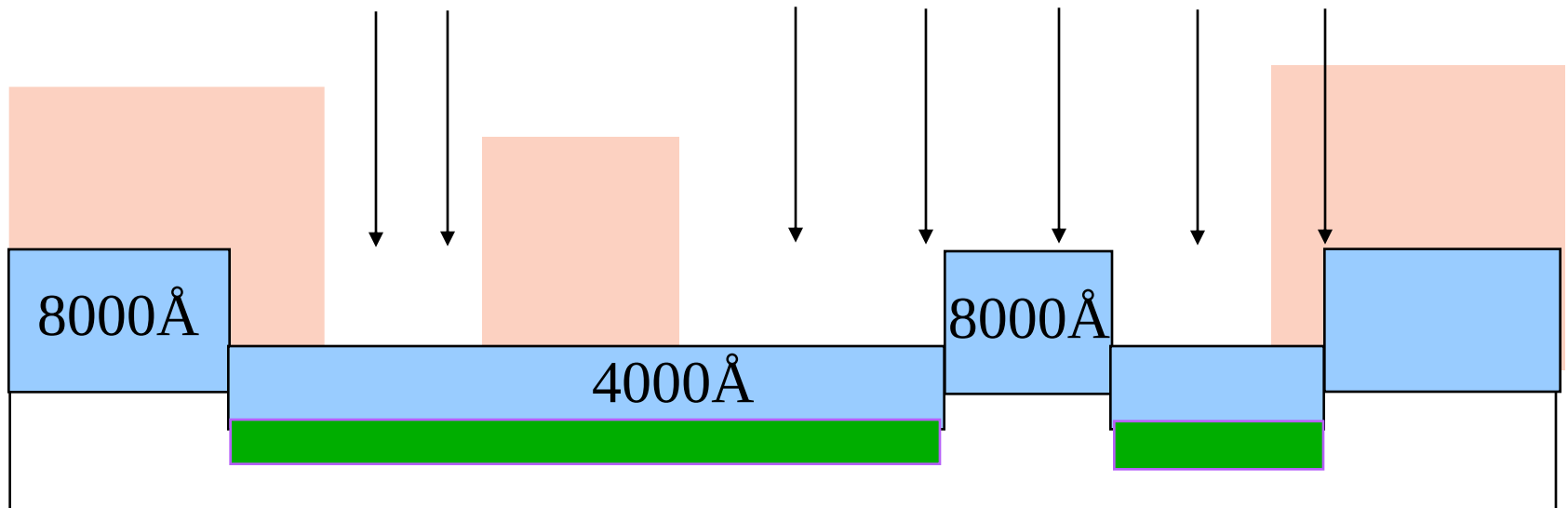
$\text{Rho} = \text{Rhos} \times t$

Tool gives Rho or Rhos depending on recipe used, automatically adjusts correction factors for wafer thickness

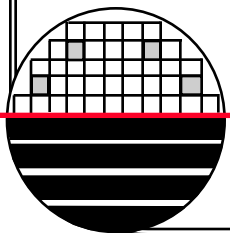


Rochester Institute of Technology  
Microelectronic Engineering

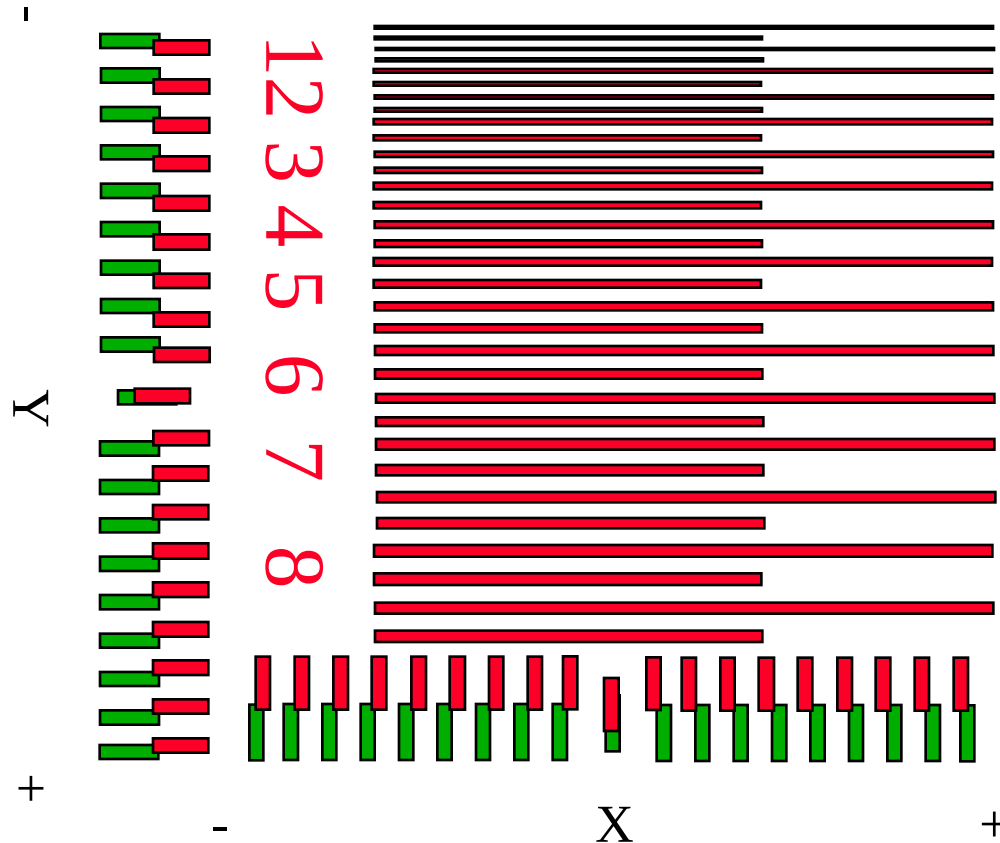


**PHOTOLITHOGRAPHY LEVEL TWO THIN OXIDE**

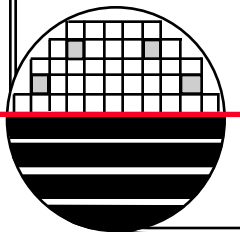
- **Coat (Recipe: COATMTL.RCP)**
  - 2000RPM for 30 seconds
  - Thickness of  $13127\text{\AA}$
- **Exposure**
  - Energy:  $250\text{mJ/cm}^2$
  - Focus:  $0.0\text{ }\mu\text{m}$
- **Develop (Recipe: DEVMTL.RCP)**
  - Wait 68 seconds
  - Hard Bake 2 min.



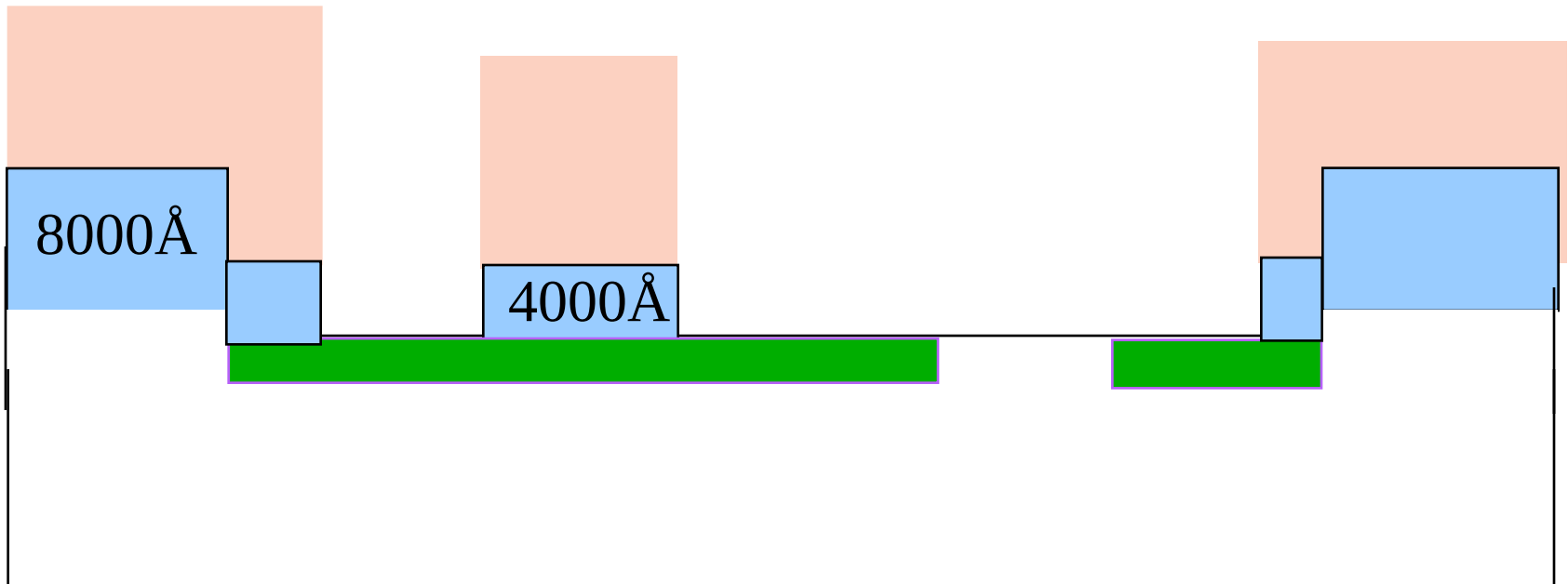
## 2<sup>ND</sup> PHOTOLITHOGRAPHY



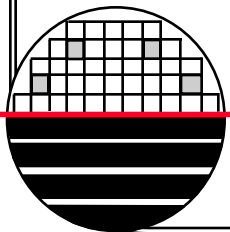
*ALIGNMENT VERNIERS*  
*CRITICAL DIMENSION (CD) STRUCTURES*



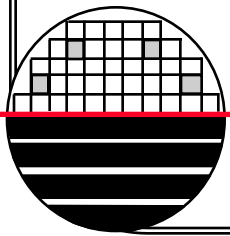
## *ETCH THIN OXIDE AREAS*



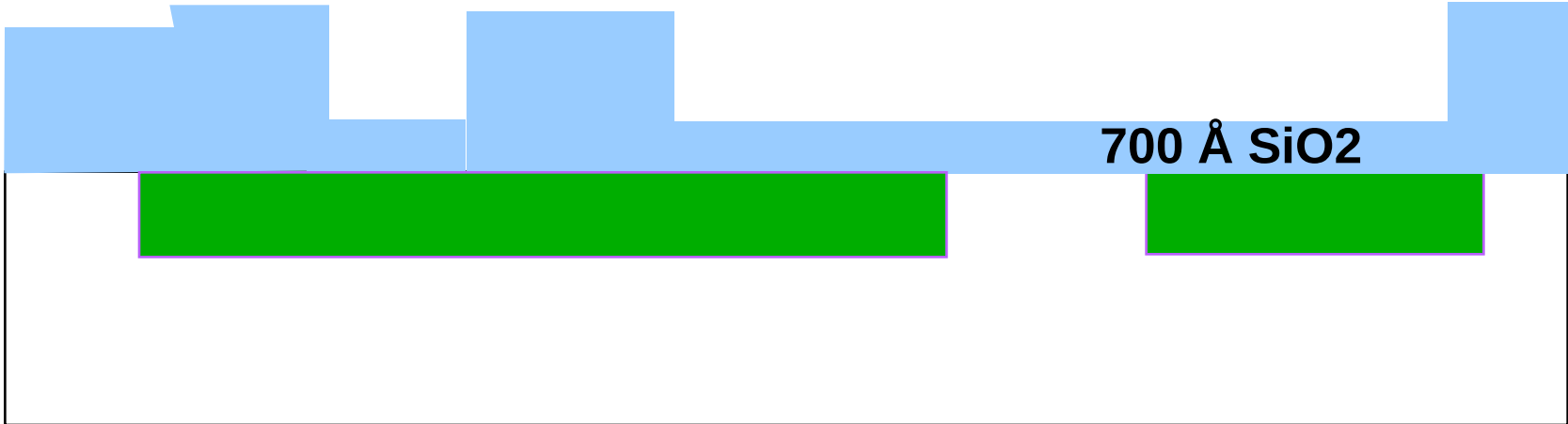
Include C2  
10 min. in 5.2:1 BOE  
20 min. in 10:1 BOE  
Rinse/Spin Dry



## *OXIDE ETCH C2 BARE*

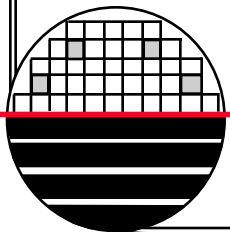


**ASH RESIST, RCA CLEAN  
GATE OXIDE GROWTH OF 700 Å**



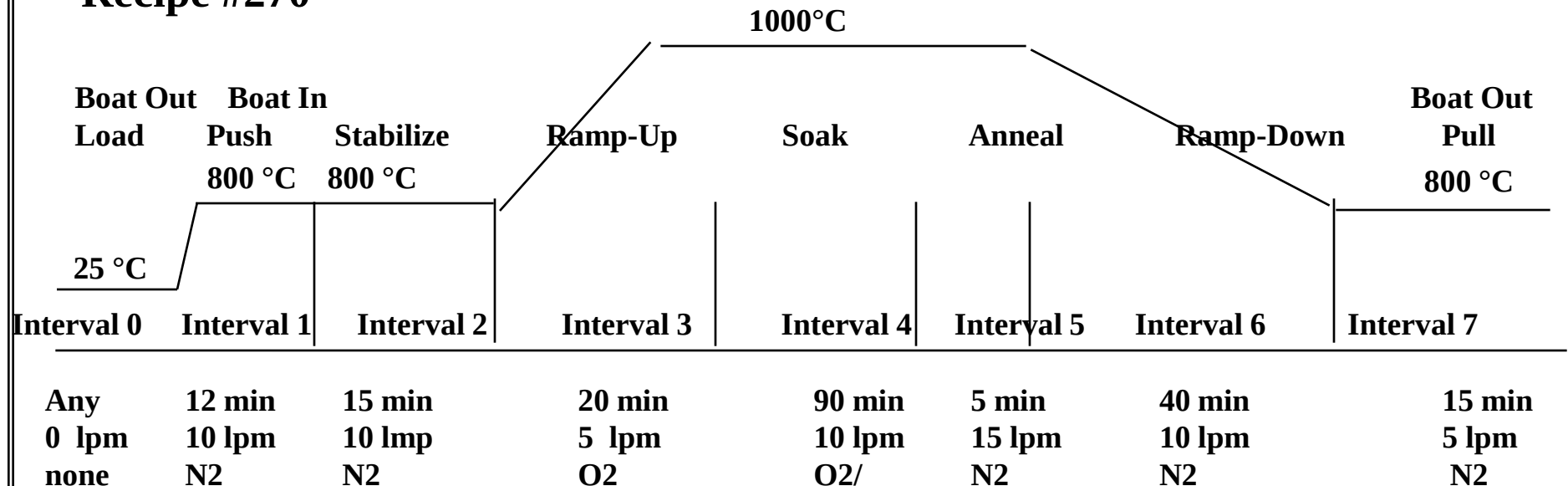
700 Å SiO<sub>2</sub>

Include C2  
Recipe 270



**BRUCE FURNACE RECIPE 270 – 700Å DRY OXIDE****Recipe #270**

Verified: 3-1-04

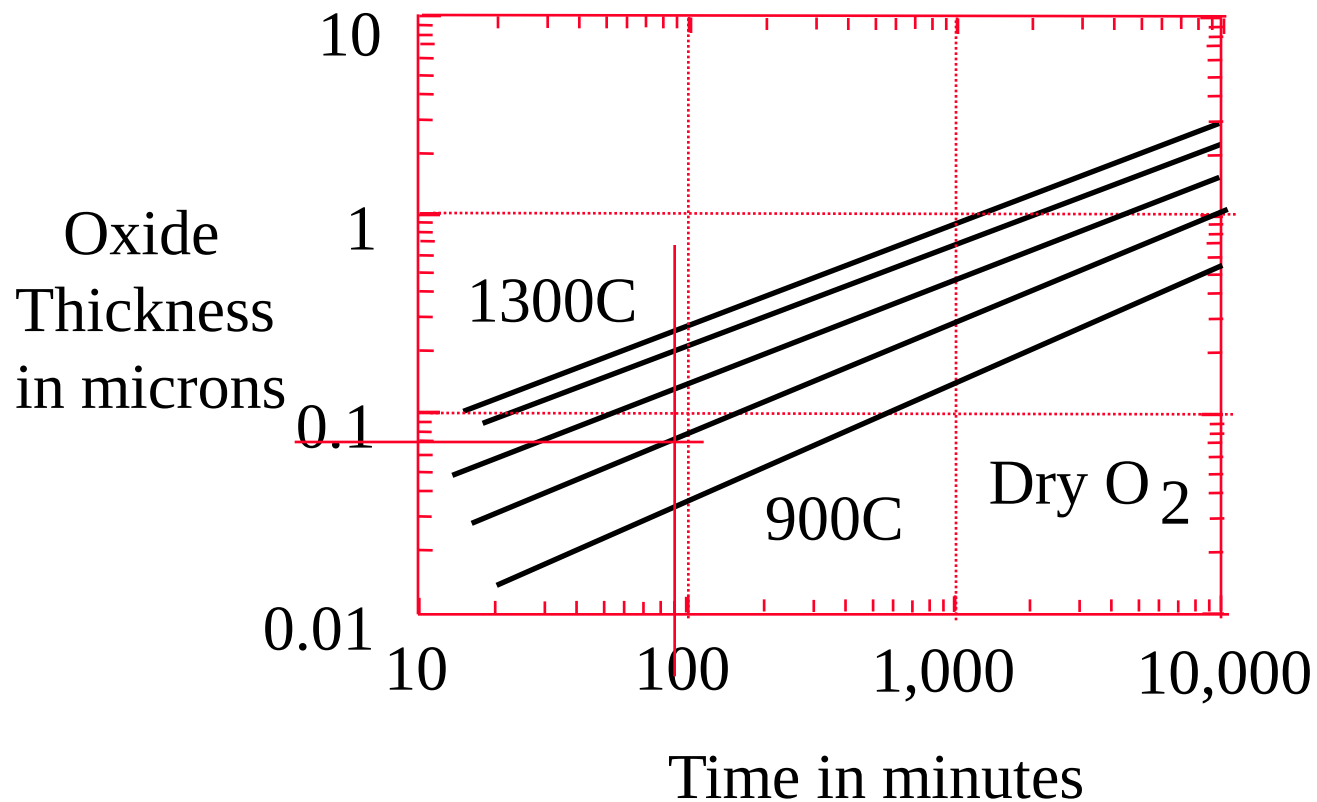


At the end of a run the furnace returns to Interval 0 which is set for boat out, 25 °C and no gas flow. The furnace waits in that state until someone aborts the current recipe or loads a new recipe.

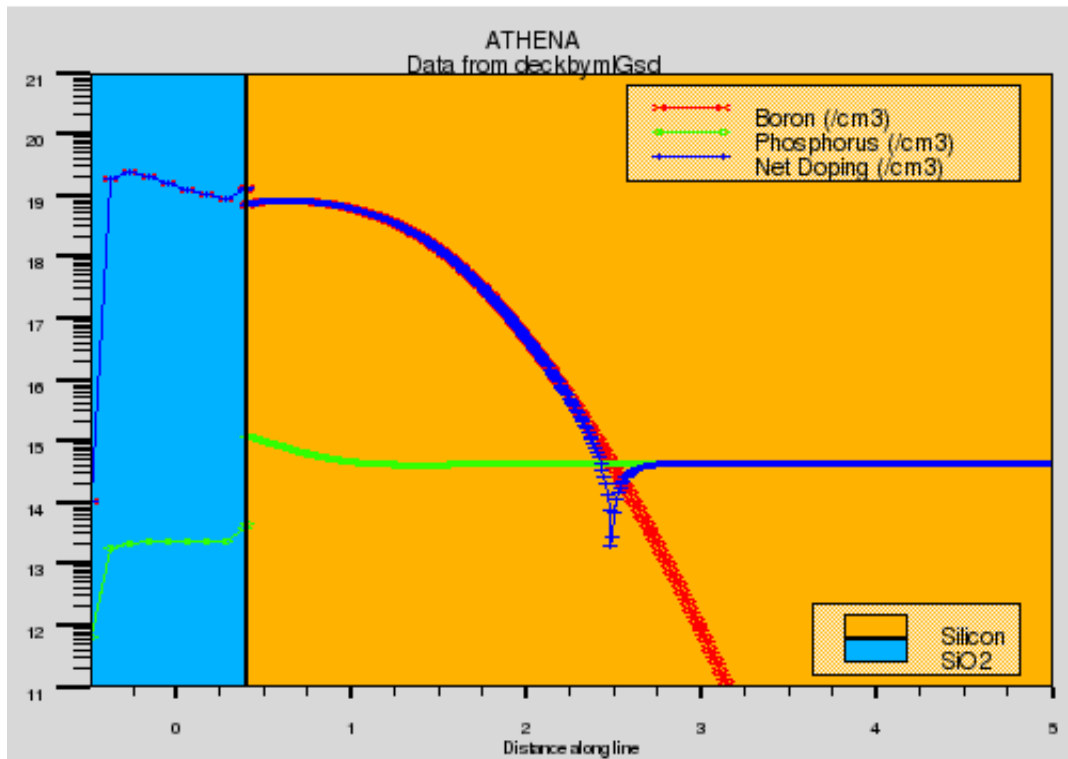
**Dry Oxide Growth, Target 700 Å**

Rochester Institute of Technology  
Microelectronic Engineering

# DRY OXIDE GROWTH CHART



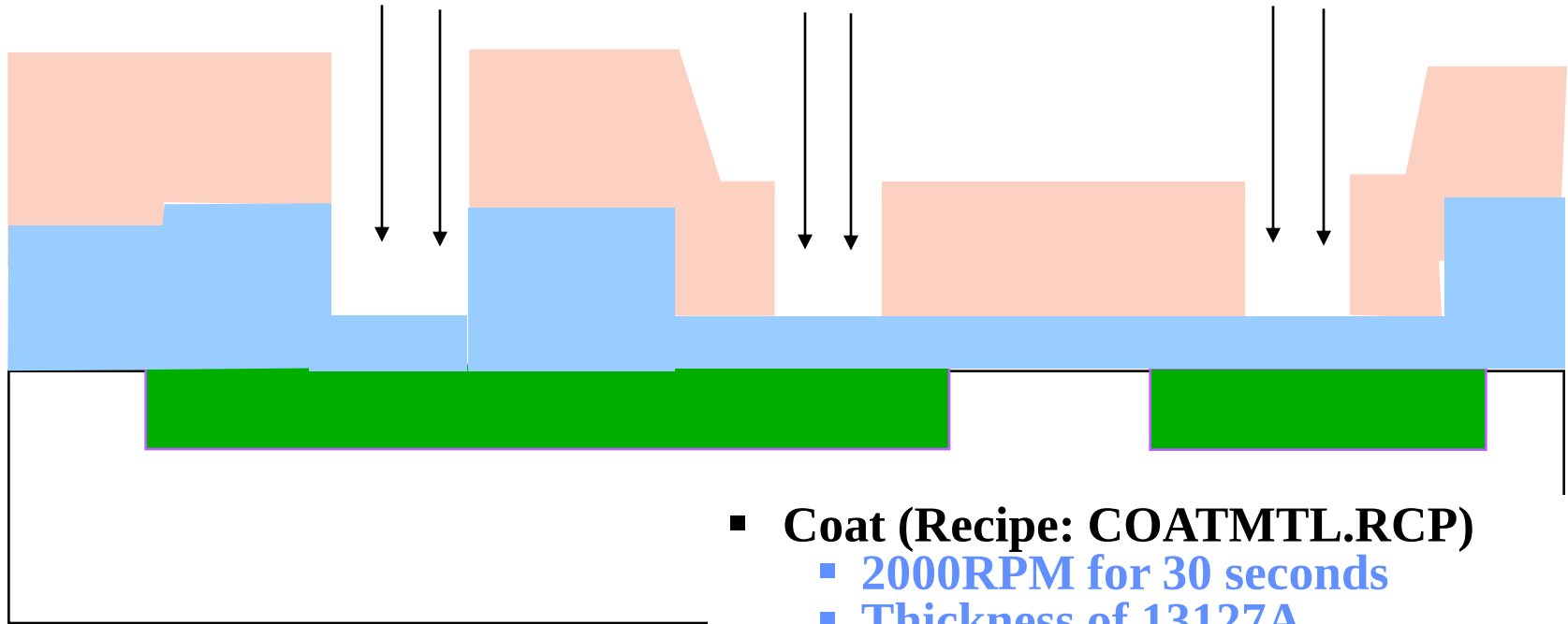
## SIMULATION AFTER ALL HIGH TEMP STEPS



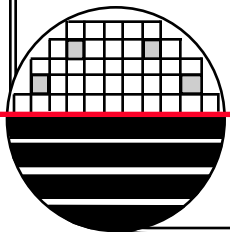
Boron  
Dose:  $2 \times 10^{15}$  ions/cm<sup>2</sup>  
Energy: 75 KeV  
 $X_j = \sim 2.0 \mu\text{m}$



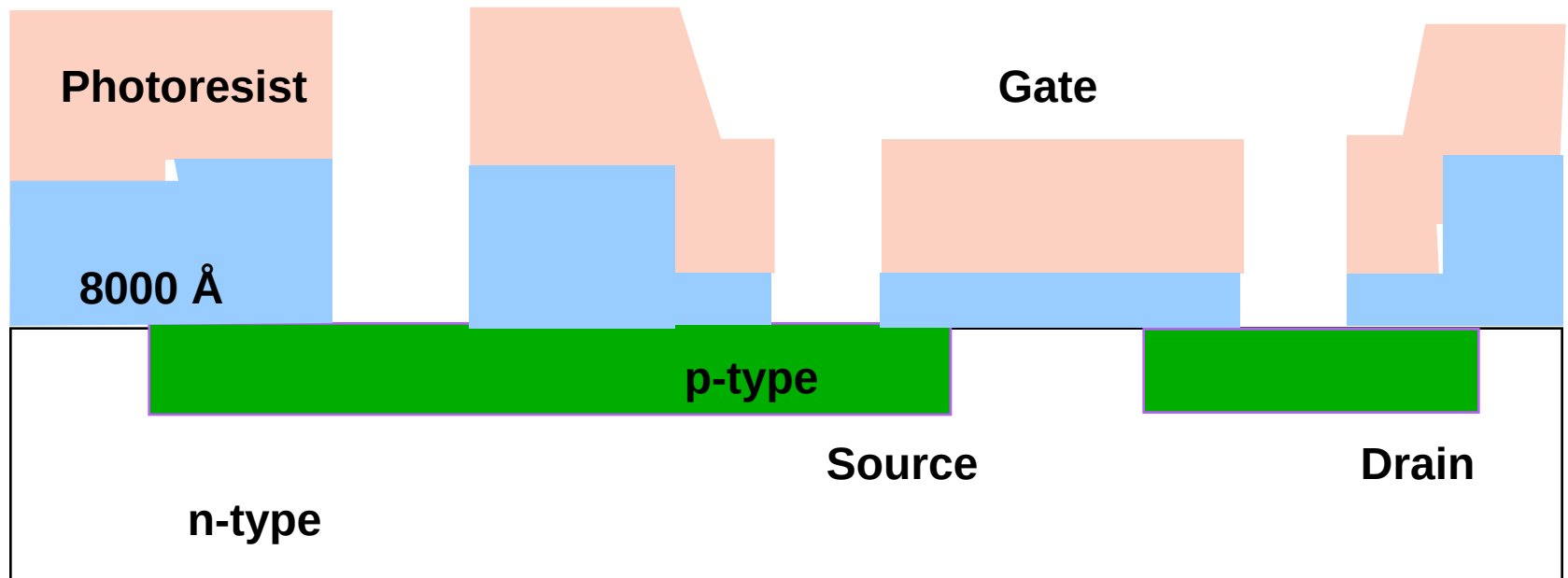
# PHOTOLITHOGRAPHY CONTACT CUT



- **Coat (Recipe: COATMTL.RCP)**
  - 2000RPM for 30 seconds
  - Thickness of 13127Å
- **Exposure**
  - Energy: 250mJ/cm<sup>2</sup>
  - Focus: 0.0um
- **Develop (Recipe: DEVCC.RCP)**
  - Wait 68 seconds
  - Hard Bake 2 min.

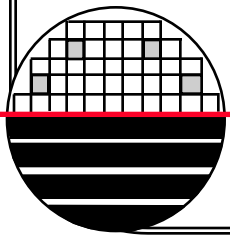


## CONTACT CUT ETCH

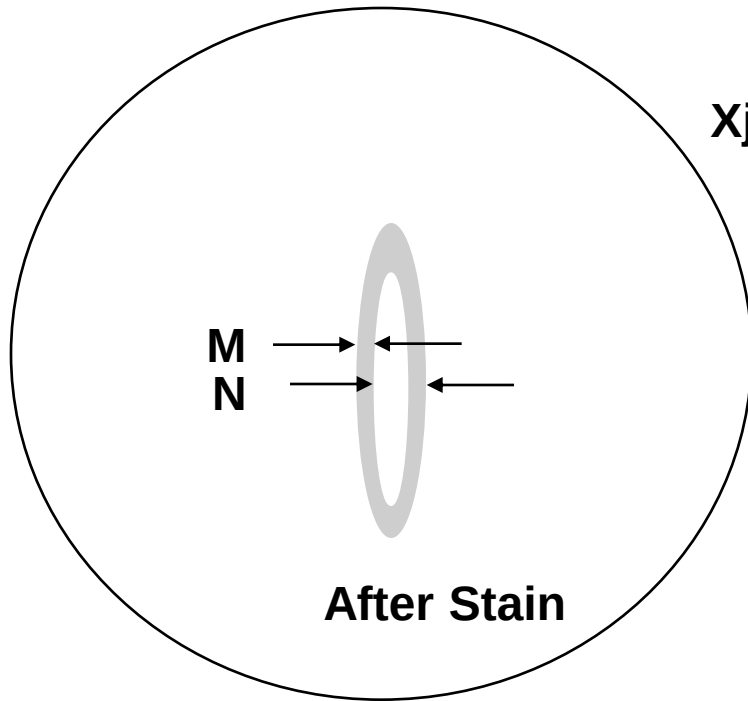


Include C2  
Be Sure to Etch for 20 min. or 10 min.  
If BOE is 10:1 then 20 min.  
If BOE is 5.2:1 then 10 min.  
SRD

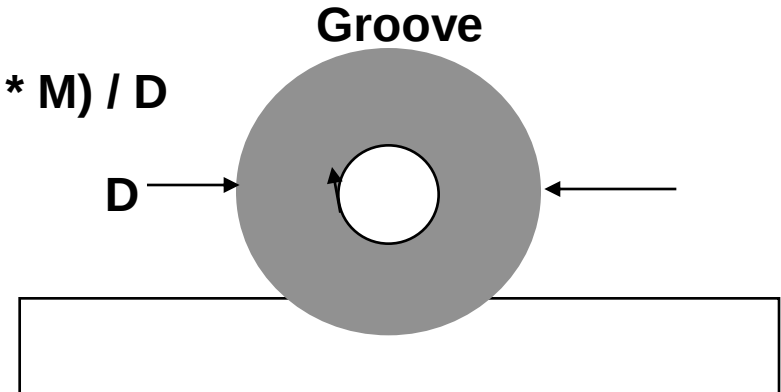
## *OXIDE ETCH C2 BARE*



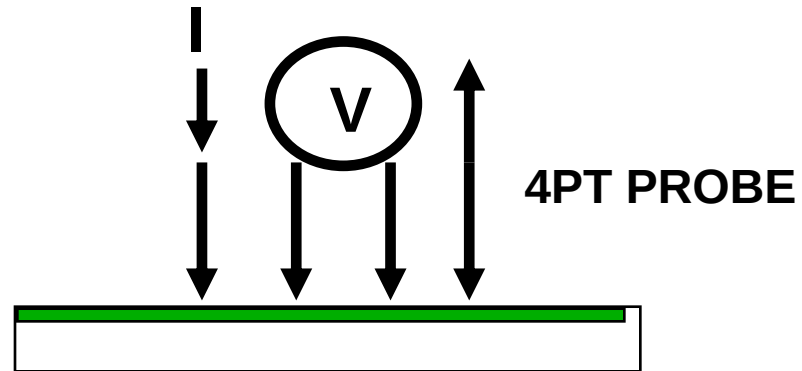
# GROOVE and STAIN and 4PT PROBE C2



$$X_j = (N * M) / D$$



p-type diffused layer



$$R_{hos} = V / I * \pi / \ln 2 \text{ ohms/square} = 4.53 V / I \text{ ohms/sq}$$

**RCA CLEAN WAFERS PRIOR TO METAL****APM**

H<sub>2</sub>O – 5000ml  
NH<sub>4</sub>OH–300ml  
H<sub>2</sub>O<sub>2</sub> – 300ml  
75 °C, 10 min.

DI water  
rinse, 5 min.

**H<sub>2</sub>O - 50**  
**HF - 1**  
**20 sec.**

**HPM**

H<sub>2</sub>O–5000ml  
HCL–300ml  
H<sub>2</sub>O<sub>2</sub> – 300ml  
75 °C, 10 min.

DI water  
rinse, 5 min.

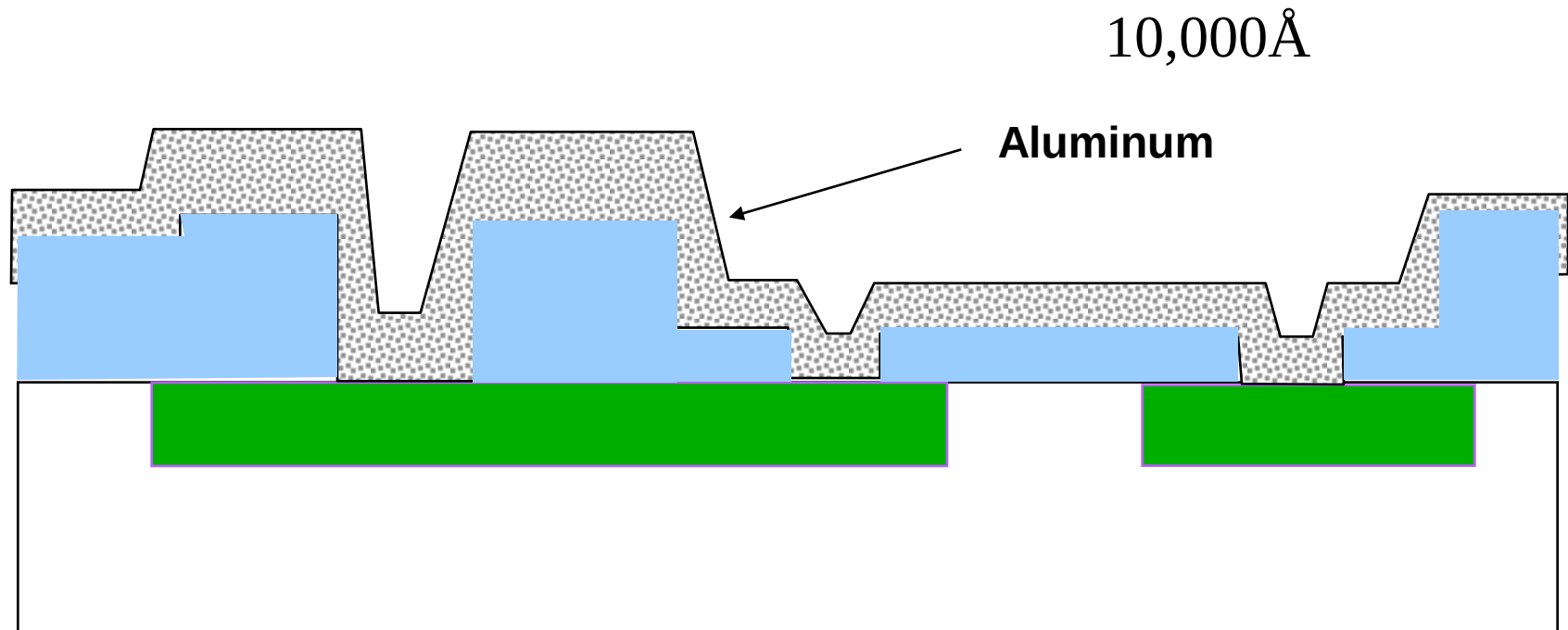
DI water  
rinse, 5 min.

**H<sub>2</sub>O - 50**  
**HF - 1**  
**20 sec.**

DI water  
rinse, 5 min.

**SPIN/RINSE**  
**DRY**

## ***ASH RESIST, RCA CLEAN and SPUTTER ALUMINUM***



Be sure to do additional dilute HF dip 10 sec at the end of the RCA clean.

## METAL DEPOSITION



### CVC 601

Thickness 10,000Å

Dep Rate ~300 Å/min

Pressure 5 mT

Ar Flow 28 sccm

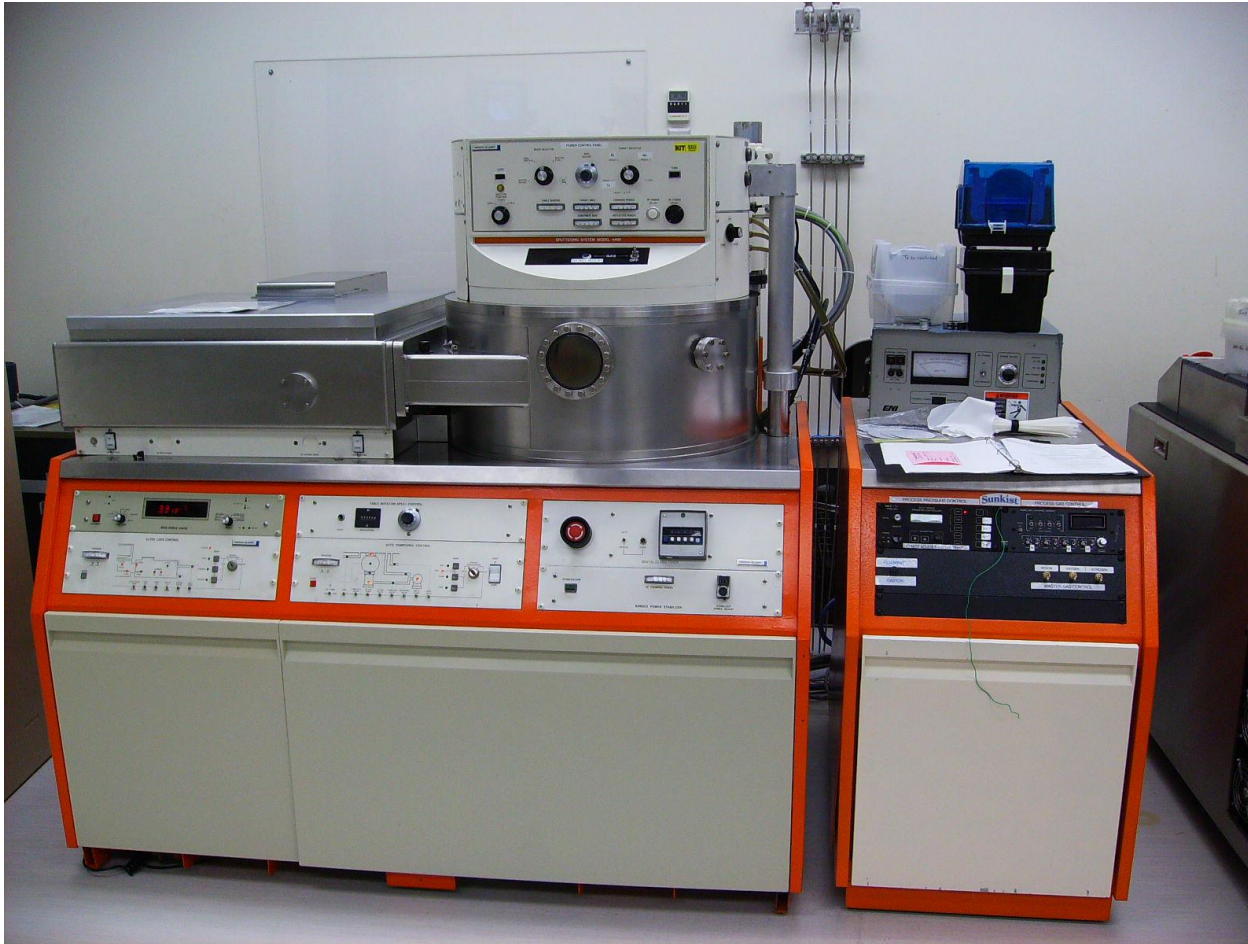
Time ~ 20 min

~2000 sec

Power 2000 watts

Pre Sputter 300 sec

# PE4400 SPUTTER / SPUTTER ETCH TOOL



*Rochester Institute of Technology  
Microelectronic Engineering*



# PE4400 – ALUMINUM DEPOSITION

## Tool Parameters

Power = 400 watts

Pressure 5 mTorr

Chiller 20 °C

Argon flow 40 sccm

Dep Rate ~64Å/min?

Time = 135 min (2hr 15min)

Blank Wafers

Table rotation speed 200 = ~6rpm

Magnet array @1° angle

Table space as close as possible

## Results

**Wafer ID: 10/4**

Ave thickness =

From P2

8665 Å

Min =

Max =

Non Uniformity =

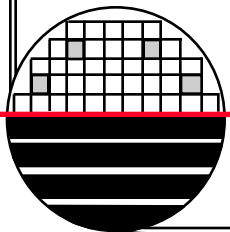
%

Surface Roughness

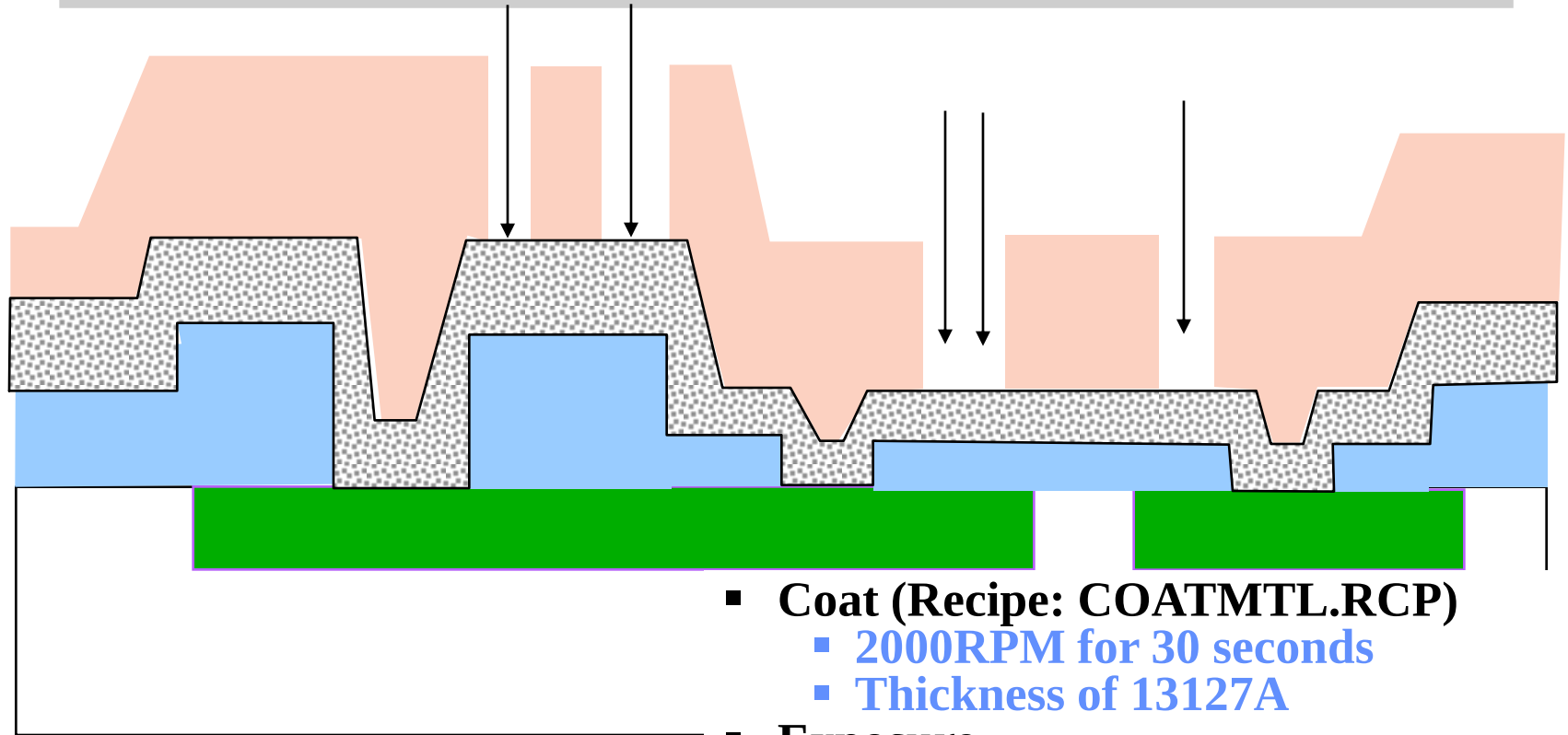
10nm great

Name:

Date:



## PHOTOLITHOGRAPHY METAL



- **Coat (Recipe: COATMTL.RCP)**
  - 2000RPM for 30 seconds
  - Thickness of 13127Å
- **Exposure**
  - Energy: 250mJ/cm<sup>2</sup>
  - Focus: 0.0 um
- **Develop (Recipe: DEVMTL.RCP)**
  - Wait 68 seconds
  - Hard Bake 2 min.

## COATMTL.RCP

### COATMTL.RCP

#### SPIN COAT

#### DEHYDRATE BAKE/ HMDS PRIMING

HMDS Vapor  
Prime  
140 °C, 60 sec.

OIR 620-10  
Resist  
Spread 400rpm  
2 sec.  
2000rpm, 30  
sec.

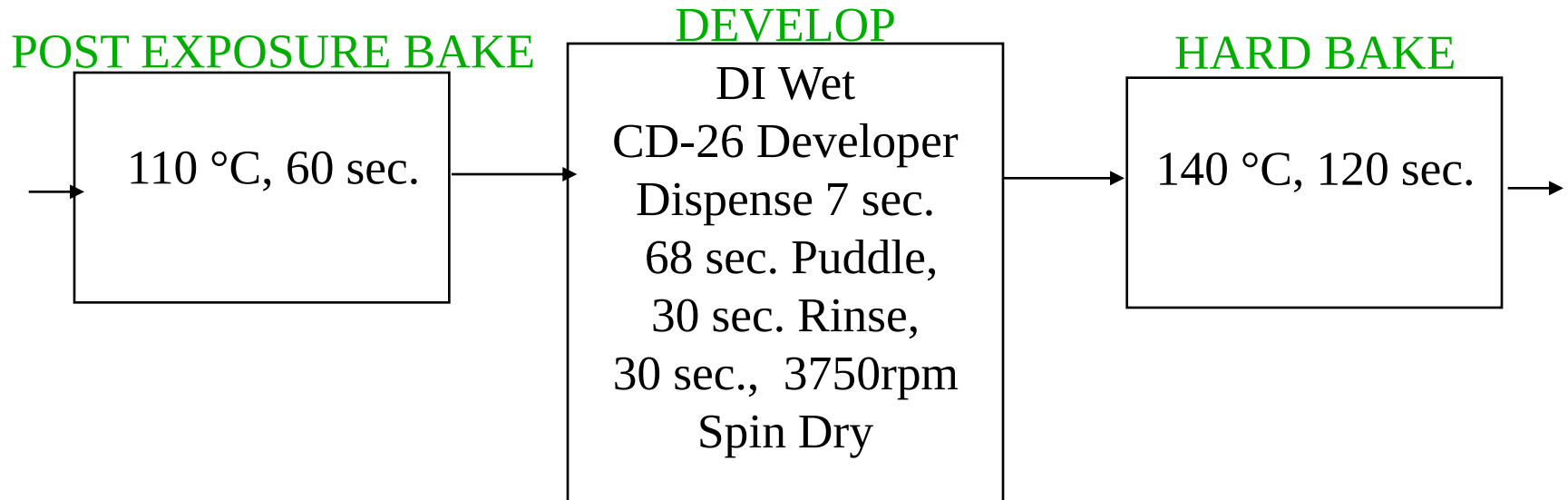
#### SOFT BAKE

90 °C  
60 sec.

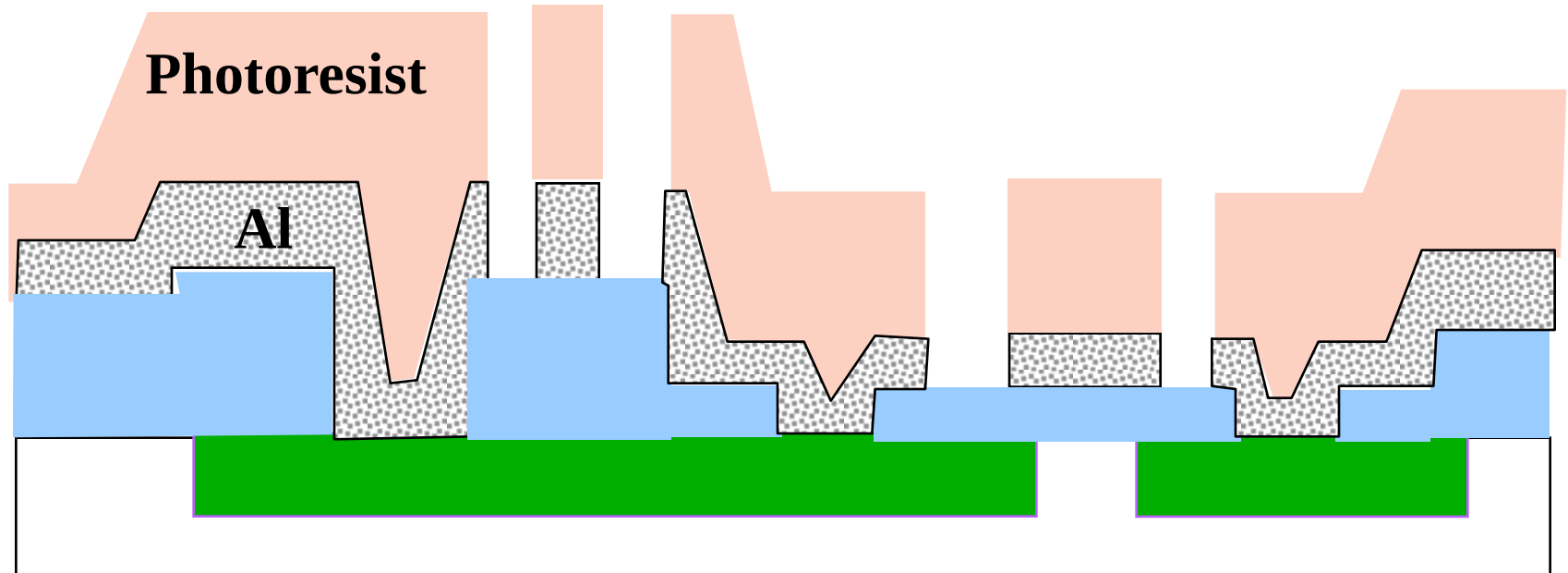
Thickness of 13,127 Å

## *DEVMTL.RCP*

### DEVMTL.RCP



## *ETCH ALUMINUM*

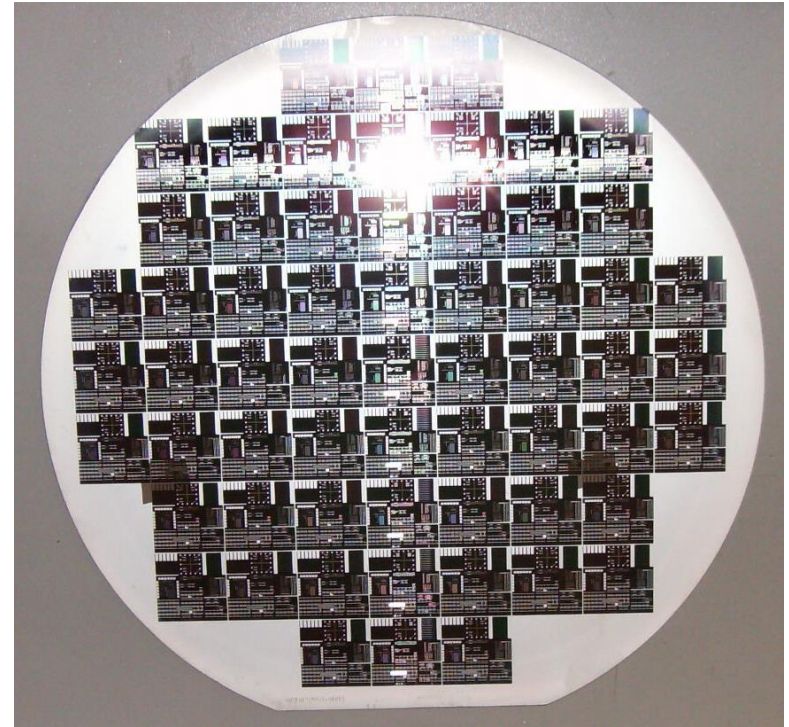


If wet etch:  
Aluminum Etch 50 °C, 3-5 min.  
DI Rinse 5 min  
Freckle Etch 2min  
DI Rinse 5 min.  
SRD Spin Dry

## ALUMINUM ETCH USING LAM4600



LAM4600



*Rochester Institute of Technology*  
*Microelectronic Engineering*

# LAM4600 ANISOTROPIC ALUMINUM ETCH

| Step       | 1     | 2    | 3    | 4     | 5    |
|------------|-------|------|------|-------|------|
| Pressure   | 100   | 100  | 100  | 100   | 0    |
| RF Top (W) | 0     | 0    | 0    | 0     | 0    |
| RF Bottom  | 0     | 250  | 125  | 125   | 0    |
| Gap (cm)   | 3     | 3    | 3    | 3     | 5.3  |
| O2 111     | 0     | 0    | 0    | 0     | 0    |
| N2 222     | 13    | 13   | 20   | 25    | 25   |
| BCl 333    | 50    | 50   | 25   | 25    | 0    |
| Cl2 444    | 10    | 10   | 30   | 23    | 0    |
| Ar 555     | 0     | 0    | 0    | 0     | 0    |
| CFORM 666  | 8     | 8    | 8    | 8     | 8    |
| Complete   | Stabl | Time | Time | Oetch | Time |
| Time (s)   | 15    | 8    | 200  | 10%   | 15   |

Rate  $\sim 38\text{\AA}/\text{s}$

Thickness =  $7500\text{\AA}$

Various tool modifications  
resulted in different etch  
rates for different years

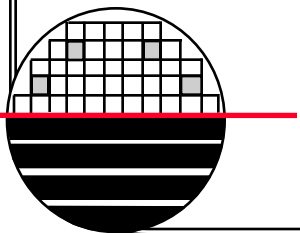
|           |      |
|-----------|------|
| Channel   | B    |
| Delay     | 130  |
| Normalize | 10 s |
| Norm Val  | 5670 |
| Trigger   | 105% |
| Slope     | +    |

Endpoint (not used)

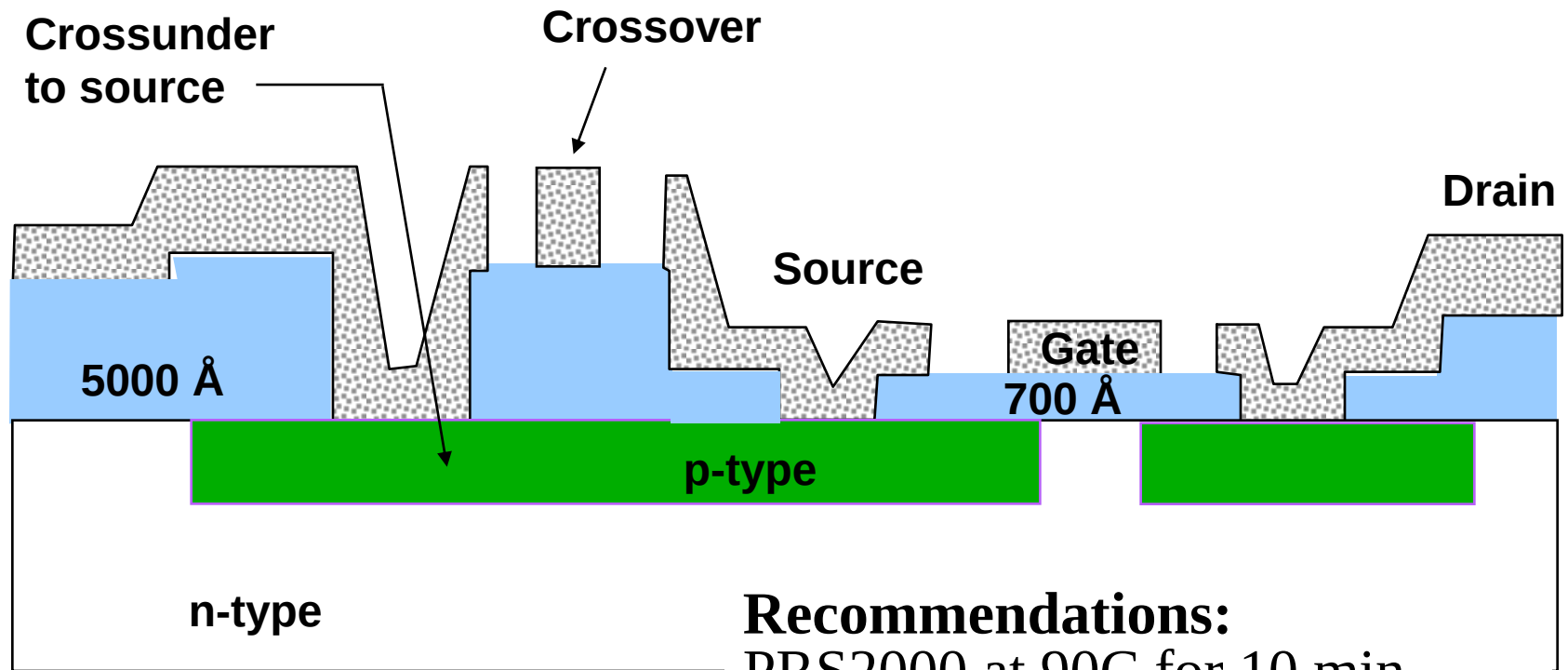
Fuller April 2013 – 200s

Fuller, January 2012 -300s

Fuller, March 2011 -230s



## ASH RESIST



### Recommendations:

PRS2000 at 90C for 10 min

Rinse 5 min. / SRD

Follow up with 6" FF on the Gas  
Sonics Asher



# BRUCE FURNACE RECIPE 101 SINTER

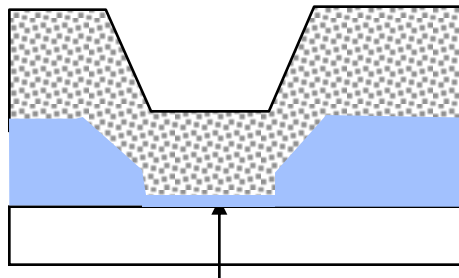
## SINTER Recipe #101

|            | Warm | Push  | Stabilize | Soak  | Anneal | Pull |       |
|------------|------|-------|-----------|-------|--------|------|-------|
|            |      |       | 400°C     |       |        |      |       |
| 25 °C      |      |       |           |       |        |      | 25 °C |
| Interval 0 | 1    | 2     | 3         | 4     | 5      | 6    |       |
| Any`       | 5    | 20    | 15        | 15    | 5      | 15   | min   |
| 0 lpm      | 10   | 10    | 10        | 5     | 10     | 5    | lpm   |
| None       | N2   | N2/H2 | N2/H2     | N2/H2 | N2     | N2   |       |

At the end of a run the furnace returns to Interval 0 which is set for boat out, 25 °C and no gas flow. The furnace waits in that state until someone aborts the current recipe or loads a new recipe.

# SINTER

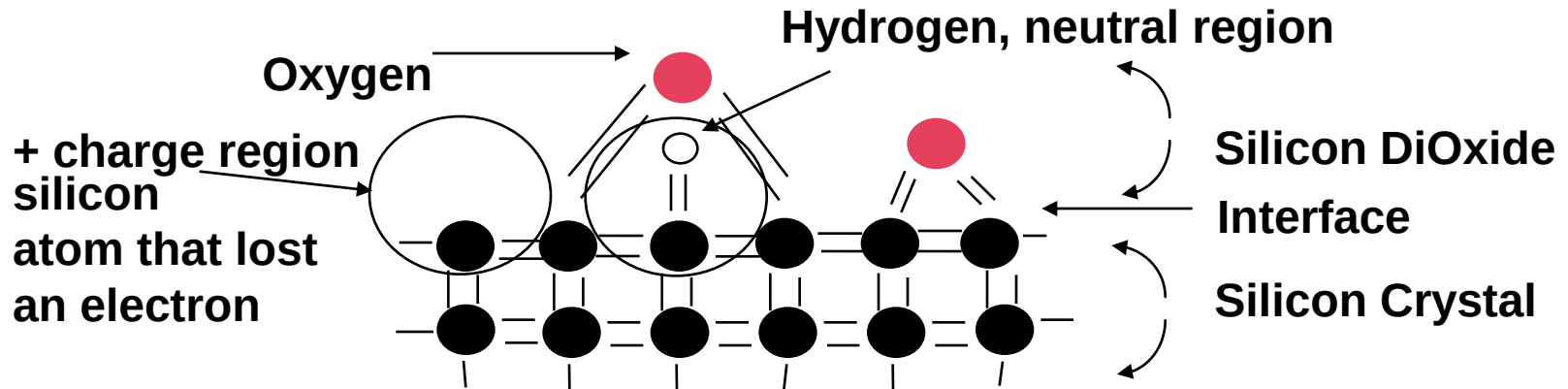
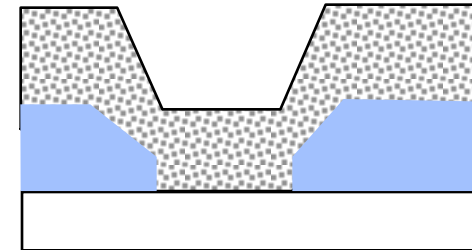
Before Sinter



Native Oxide

**Reduce Contact Resistance**

After Sinter



**Reduce Surface States**

*TEST*

X

D

D

S

G

DRAIN

GATE

SOURCE

CROSSUNDER

PMOS TRANSISTOR

## TEST RESULTS

